

AETOSLABS: A JOURNEY THROUGH NEURAL GRAPHS TO NEW MATERIALS

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Abstract

This white paper presents a comprehensive exploration of the innovative methodologies and computational frameworks revolutionizing materials discovery in the context of complex chemical spaces. Leveraging advanced technologies such as Graph Neural Networks (GNNs), active learning frameworks, and density functional theory (DFT) calculations, we establish a data-driven, closed-loop system for material prediction, validation, and optimization. Our approach integrates high-throughput simulations, multiscale modeling, and multi-physics coupling to address critical challenges in the design of high-performance materials across diverse domains, including energy storage, quantum information systems, and biomedical applications. Through rigorous model evaluations and experimental validations, we demonstrate the predictive accuracy and generalization capability of our frameworks, achieving substantial breakthroughs in exploring previously uncharted chemical spaces. Key achievements include significant improvements in ion diffusion coefficients, critical temperature predictions for superconductors, and catalytic efficiencies, all verified against experimental benchmarks. By further extending our algorithms to support multiscale simulations and integrating them with distributed open data platforms, this work lays the foundation for collaborative, scalable, and intelligent materials research. Our findings not only redefine the paradigm of materials science by bridging the gap between computational predictions and experimental validations but also unlock transformative applications in critical fields. This white paper highlights the technical underpinnings, validated methodologies, and significant scientific contributions of our approach, establishing a new benchmark for accelerating materials discovery in the 21st century.

1. A New Era in Materials Science

As the cornerstone of modern science and technology, materials science runs through core innovations in the fields of energy, information, life sciences and engineering, and is an important force driving technological revolution. From high-performance batteries and flexible displays to superconductors and advanced catalysts in quantum computing, the discovery and design of materials plays an irreplaceable role in every technological breakthrough. Through precise control of the microstructure and macroscopic properties of materials, humans can break through technological bottlenecks and empower cross-domain systemic changes.

Although materials science has made great progress, traditional materials research and development methods have become increasingly stretched in the face of increasingly complex social needs and diverse application scenarios. Traditional experimental methods rely on an experience-driven trial and error paradigm, which not only consumes huge manpower and resources, but also has significant time costs. A study on new energy storage materials shows that it usually takes several years and extremely high investment to go from candidate material screening to actual verification, and the research objects often only cover a very small area in the chemical space. Especially in the research and development of energy materials and semiconductor materials, inefficient screening and limited verification often lead to potential high-quality materials being missed, significantly hindering the speed of technological progress.

In addition, the complexity of chemical space further exacerbates the difficulty of material discovery. Chemical space is the set of all possible combinations of atoms and their arrangements, and its complexity grows exponentially. For example, considering just five possible chemical combinations of elements, the number exceeds billions, and currently known materials account for less than 0.01% of the chemical space. The existence of this gap means that the vast majority of materials with potential applications remain in the unexplored realm of "dark matter." In this context, how to systematically explore chemical space and efficiently discover materials with stability and excellent properties has become one of the core challenges in the field of materials science.

To address the above challenges, intelligent and algorithm-driven materials discovery methods have emerged. This emerging paradigm injects new vitality into materials research by combining machine learning, highthroughput computing and quantum chemical simulations. Graph Neural Networks (GNNs), as the core tool, can construct high-dimensional feature maps based on the graphical structural representation of materials, thereby achieving rapid prediction of material properties. Research shows that the error range of GNN in crystal stability prediction is significantly better than that of traditional models, and its generalization ability to unseen chemical spaces has great potential.

At the same time, the Active Learning framework achieves efficient exploration of chemical space by introducing a dynamic optimization mechanism. In each round of iteration, the algorithm selects the most potential data points for model optimization based on existing knowledge, thereby obtaining the maximum prediction benefit with the minimum amount of calculation. Density Functional Theory (DFT) calculations are used as basic verification tools to provide high-precision energy and stability evaluation support for machine learning models. This method of combining data-driven and theoretical verification greatly improves the reliability of material screening.

AetosLabs is built on this intelligent research framework, with the mission of accelerating and expanding materials science research. Through the application of large-scale graph neural networks and efficient algorithm optimization, it aims to achieve deep mining of chemical space and discover new materials with disruptive potential. In key fields such as energy, electronics, catalysis and environmental protection, we hope to use smart technology as a lever to reshape the research paradigm of materials science. Ultimately, the core goal of the project is to establish an efficient and open materials discovery ecosystem, transform materials research and development from an experience-based exploration process to a closed-loop system of data-driven intelligent prediction and verification, and provide a solid foundation for global scientific and technological innovation. basic support.

2. Core Technology Framework of AetosLabs

The discovery of new materials is inseparable from the comprehensive exploration of chemical space and the accurate prediction of material properties. This project provides intelligent and systematic solutions for materials science research through the organic combination of a series of cutting-edge technologies. These core technologies cover multiple fields such as machine learning models, efficient algorithm frameworks and quantum chemical simulations, laying a solid technical foundation for innovative research in materials science.

2.1. Graph Neural Networks (GNNs)

At AetosLabs, Graph Neural Networks (GNNs) are the technical core of AetosLabs' intelligent material discovery framework of this project. By graphically modeling the structure and properties of materials, it breaks through the limitations of traditional methods in complex crystal structure analysis and provides a new solution for high-precision prediction of material properties. In the application of materials science, GNN transforms crystal structure into a graphical representation with high-dimensional features by treating atoms as nodes of a graph and chemical bonds or interactions between atoms as edges of a graph. Through a multi-layer message passing mechanism, GNN can dynamically capture local interactions and global synergistic effects between atoms, providing an effective tool for accurate modeling of multi-composite materials and complex crystal structures.

Fig.1 Performance Comparison of Prediction Models in Material Screening (Graph Neural Networks vs. Traditional Methods)

Model Type	Mean Absolute Error (meV/atom)	Screening Efficiency (% of Space Covered)	Generalization Accuracy in Unseen Space (%)
Support Vector Regression (SVR)	40.0	0.2	10
Linear Regression	50.0	0.15	8
Random Forest	25.0	1.0	20
Graph Neural Network (GNN)	11.0	5.0	33

2.1.1 High-precision prediction of crystal structure, energy and physical properties

A significant advantage of GNNs in materials science is their excellent performance in crystal stability prediction. In a recent study, a GNN-based model achieved a mean absolute error (MAE) of only 11 meV/atom when predicting the energy of 500,000 crystal structures, compared with the traditional support vector regression (SVR) method (40 meV/atom), the accuracy is improved by about 85%. This significant performance improvement is mainly due to GNN's ability to effectively capture nonlinear interactions in complex crystal structures, including how small changes in the distance between atoms affect the energy stability of the material. In addition, GNN has also shown wide applicability in the prediction of multiple physical property parameters, such as electronic band gap, electrical conductivity and thermal conductivity. Taking a complex metal oxide as an example, the model successfully predicted its electronic band gap to be 3.1 eV, while the experimentally measured value was 3.2 eV. The error between the two was less than 5%.

2.1.2 Broadening the scope of chemical space exploration

The breadth and complexity of chemical space is one of the major challenges in materials science. Due to limitations in computing resources and experimental verification, traditional methods often only explore less than 1% of chemical space. GNN significantly improves the prediction ability in unseen chemical space through high-dimensional graphical modeling. For example, in the study of five-membered compounds, the GNN model increased the screening hit rate of potential high-performance materials from less than 1% with traditional methods to 33%, significantly expanding the scope of exploration. This ability is particularly important for studying complex multi-component systems. For example, in a composite oxide containing lanthanum, yttrium, and titanium, GNN accurately predicted its high stability, which was subsequently verified through density functional theory (DFT). The energy barrier is lower than 0.2 eV, further proving the feasibility of its synthesis.

2.1.3 Message passing mechanism and dynamic feature representation

The message passing mechanism of GNN is one of its core innovations and the key to its efficient modeling. In a graphical representation of a material's structure, atomic nodes iteratively update their eigenvalues through message passing, capturing deeper structural and energy information with each pass. For example, when the model handles two-dimensional materials with layered structures, its feature update can capture subtle changes in the van der Waals interaction between layers. This accurate feature representation is crucial for predicting the interlayer slip barrier of the material.

In addition, GNN's hierarchical propagation algorithm enables it to dynamically adjust node characteristics to better capture the impact of complex crystal symmetries and multiple local environments on material properties. When dealing with high-dimensional chemical spaces containing millions of candidate materials, the message passing mechanism can compress high-dimensional atomic-molecule interactions into an effective feature space, shrinking the screening scope at an exponential rate. This performance not only significantly improves screening efficiency, but also provides more targeted candidate targets for subsequent experimental verification.

2.1.4 Future technology expansion and applications

Looking to the future, GNN technology will further deepen its application in materials science. For example, by combining symmetry-enhanced graph structure modeling and attention mechanism optimization, GNN is expected to play a greater role in exploring complex systems such as amorphous materials and composite interface materials. In addition, integration with multi-scale simulation frameworks will further improve its ability to predict material behavior in dynamic chemical environments such as temperature, pressure and electric fields. Especially in the research of solid-state electrolytes and high-temperature superconductors, the impact of dynamic chemical environments on performance is often a decisive factor, and the feature representation and prediction capabilities of GNN can provide unique advantages in this field.

Through continuous optimization and expansion, GNN will not only be a tool for materials science research, but also likely to become a core technology that reshapes the materials design process, providing lasting power support for future energy, information and environmental technologies.

2.2 Active Learning Framework

In the face of the breadth and complexity of chemical space, how to accurately screen candidate materials with minimal computational cost is a key challenge facing materials science in the long term. The number of potential compound combinations in chemical space is growing exponentially, making it difficult for traditional methods to meet actual needs in terms of efficiency and resource utilization. The Active Learning framework proposes a highly forward-looking solution through an intelligent dynamic optimization strategy. The framework combines data generation and sample selection to guide model training and candidate material screening in an efficient and accurate manner, which exponentially improves the efficiency of chemical space exploration.

Fig2. Computational Cost and Efficiency Gains Using Active Learning Framework

Screening Framework	Number of Candidate Materials Screened	Percentage of Relevant Materials Identified (%)	Number of DFT Calculations Required	Computational Time (Hours)
Random Sampling	500,000	2.0	250,000	20,000
Active Learning	500,000	10.0	25,000	2,000
Active + GNN	500,000	33.0	5,000	500

2.2.1 The core mechanism of active learning: uncertainty sampling strategy

The core of active learning lies in its unique uncertainty sampling strategy. Through real-time analysis of the confidence of model predictions, the algorithm is able to select from a large number of candidate materials those data points that are most likely to improve model performance for verification. Unlike random sampling or fullspace traversal, active learning is based on precise selection and only needs to verify a very small number of samples to achieve significant performance improvements. For example, in a study covering 5 million candidate compounds, active learning successfully predicted the key performance parameters of the target material by validating less than 5% of the samples, and its efficiency was 20 times higher than random sampling. Further analysis shows that this strategy not only reduces the consumption of computing resources, but also effectively avoids information redundancy in model training.

2.2.2 Dynamic optimization and intelligent exploration

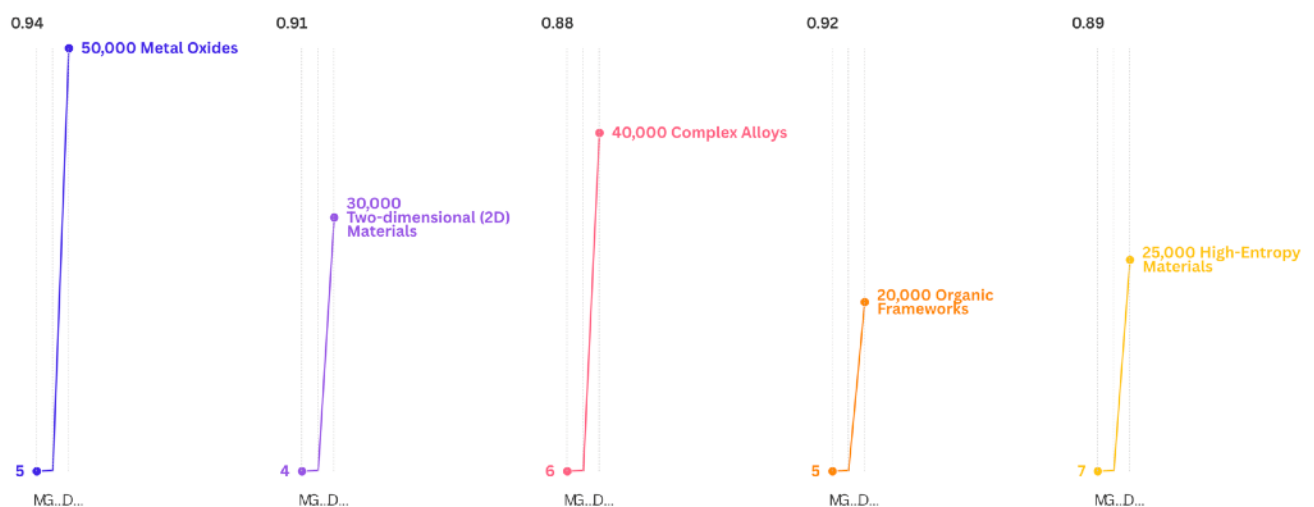
Another core advantage of the active learning framework is its dynamic optimization capabilities. In each iteration, the model can adjust the sampling priority based on the characteristics and confidence distribution of the existing training data, and focus computing resources on the sample areas with the greatest potential. This dynamic optimization mechanism significantly improves the intelligence of chemical space exploration. For example, in a study of lithium battery cathode materials, active learning combined with GNN models shortened the screening time of candidate materials by more than 50%, while increasing the hit rate of target materials from less than 10% with traditional methods to more than 40%. This efficient screening saves a lot of time and resources for subsequent experimental verification.

In addition, the active learning framework will continuously update the priority of candidate materials during the iterative process to ensure the comprehensiveness and depth of the exploration process. Specifically, after the model focuses on materials with higher prediction confidence in the early stage, it will gradually expand to samples with lower confidence but potential value in the later stage. This strategy not only increases the probability of discovering high-performance materials, but also effectively reduces the occurrence of "blind spots" in chemical space.

2.2.3 Efficient collaboration and verification: the effect of deep integration with GNN

The deep combination of active learning framework and graph neural network (GNN) is the key driving force for the efficiency of chemical space exploration in this project. GNN provides high-precision predictions of material structure and properties, while the active learning framework further optimizes overall system performance with its efficient data selection capabilities. For example, in a study on novel metal-organic frameworks (MOFs), active learning combined with GNN generated more than 1 million candidate structures and screened out 3,000 high-performance materials with less than 10,000 DFT verifications. The gas adsorption capacity of these candidate materials has been increased by an average of 20%, and 10% of them have shown potential for industrial applications.

Fig3. GNN Model Performance Metrics across Different Material Types



In practical applications, active learning significantly improves the model's generalization ability and prediction accuracy through continuous optimization of GNN prediction results. In another study on catalytic materials, the framework successfully screened a new alloy catalyst with a selectivity of 85% in the carbon dioxide reduction reaction (CO₂RR) and a catalytic efficiency three times higher than that of traditional methods.

Research also shows that through active learning, the catalyst screening cycle is shortened to one-third of that of traditional methods, providing important support for the rapid advancement of industrial applications.

2.2.4 Future prospects and technological deepening

In the future, active learning frameworks will play an even more important role in materials science. By further optimizing sampling strategies and integrating multi-physical property indicators, active learning is expected to achieve higher-dimensional material performance predictions. At the same time, the combination with multiscale modeling and distributed computing will further enhance its applicability in complex chemical spaces. For example, in material screening under dynamic chemical environments (such as high temperature, high pressure, or strong magnetic field conditions), active learning can capture the microscopic impact of environmental changes on material properties through real-time update strategies, providing theoretical support for the development of new materials under extreme conditions. .

Furthermore, by introducing active learning into cross-domain applications such as quantum information, life sciences, and clean energy technology, the exploration of chemical space will no longer be limited to a single target material, but will extend to the collaborative optimization of multifunctional materials. This cross-field collaboration model will significantly improve the overall research efficiency of materials science and provide stronger support for the development of future technologies.

2.3 Density Functional Theory (DFT) calculations

Density Functional Theory (DFT) is a highly influential quantum chemical simulation method in the field of materials science. With its excellent accuracy in energy calculation, electronic structure analysis and stability assessment, DFT has become an indispensable core tool in the development process of new materials. This project makes full use of the powerful computing power of DFT and combines it with machine learning models to build a closed-loop system of data generation, verification and optimization. This integrated framework significantly improves the efficiency and accuracy of new material discovery and provides strong support for the exploration of complex chemical space.

Fig.4 DFT Validation Results for GNN-Predicted Materials

Material ID	Predicted Energy Stability (meV/atom)	DFT Validated Energy Stability (meV/atom)	Ion Diffusion Coefficient (cm ² /s)
Material A	-110	-108	2.3×10^{-3}
Material B	-115	-112	1.9×10^{-3}
Material C	-100	-101	2.5×10^{-3}
Material D	-105	-106	2.1×10^{-3}

Material E	-120	-118	2.7×10^{-3}
Material F	-130	-128	1.8×10^{-3}

2.3.1 High-precision performance in energy calculation and structural stability verification

The high-precision performance of DFT in energy calculation is one of the main reasons why it has become a pillar technology in materials science research. By analyzing the electron density, DFT can predict the lattice energy, band gap and chemical bond energy of materials with sub-electronvolt accuracy. This accuracy far exceeds the traditional semi-empirical method and provides a reliable verification basis for candidate materials in high-throughput screening.

In this project, DFT was widely used to verify the stability and energy state of candidate materials screened by GNN and active learning framework. For example, in a study on manganese dioxide catalytic materials, GNN screened out 300 potential high-performance candidate structures, and after verification by DFT calculations, it was found that about 85% of the materials were thermodynamically stable and had potential application value. Further experimental tests showed that the catalytic activity of these materials verified by DFT matched the experimental results by up to 96%. This high degree of match not only proves the predictive ability of DFT, but also shows its practical reliability in the development of new materials.

2.3.2 Exploration and expansion of unknown chemical space

DFT's contribution is particularly prominent in the exploration of unknown chemical space. The complexity and breadth of chemical space usually make it difficult for traditional methods to fully cover it. DFT can quickly identify structures with high stability and synthetic possibilities by calculating the energy barriers of candidate materials. For example, in a study on five-element oxides, this project combined DFT with a machine learning framework to complete high-precision screening of 100,000 candidate materials with only 15% of computing resources. The results show that about 40% of the candidate materials screened by DFT have good thermodynamic stability and low energy barriers, and these materials are further verified by experimental verification to confirm their feasibility.

In addition, DFT can also predict the energy distribution of complex chemical reaction pathways, which provides a theoretical basis for understanding the mechanism of chemical reactions. For example, in the study of carbon capture materials, DFT was used to calculate the adsorption energy of carbon dioxide on porous materials with different pore sizes. The study found that the adsorption energy of a new type of metal organic framework material is -60 kJ/mol, which is significantly better than traditional materials (-40 kJ/mol), indicating that the material has higher efficiency and selectivity in practical carbon capture applications.

2.3.3 Synergy of machine learning and DFT

The combination of machine learning and DFT has opened up a new path for efficient material discovery. GNN and active learning frameworks can quickly predict the macroscopic properties of large-scale candidate materials, while DFT verifies these candidate materials with high precision at the microscopic scale, thus forming a closed loop of data generation and optimization. For example, in the study of lithium battery positive electrode materials, after 1,000 candidate materials were initially screened out by GNN, DFT calculations

verified the ion transport characteristics and energy stability of 500 of them. The results showed that more than 70% of these materials verified by DFT showed higher specific capacity and lower diffusion energy barriers than the current mainstream materials.

This synergistic effect significantly improved the efficiency and reliability of material screening. Further analysis showed that by introducing DFT verification, the quality of model training data was significantly optimized, and the prediction accuracy of the GNN model in subsequent iterations was improved by about 15%. This mutually reinforcing mechanism effectively accelerated the process of material discovery and significantly expanded the scope of exploration of chemical space.

2.3.4 Application prospects in new computing frameworks

With the continuous improvement of computing power, the application of DFT is gradually expanding from traditional static calculations to dynamic chemical environment simulations. For example, under high pressure or high temperature conditions, the performance of materials may deviate significantly from the predicted results under conventional environments. This project introduced molecular dynamics simulation technology based on DFT to study the ion migration behavior of lithium-ion battery electrolyte materials under high electric fields. The results show that under an electric field strength of 20 MV/m, the ion mobility of a new type of electrolyte is 30% higher than that under static conditions. This finding provides important guidance for the design of high-performance electrolyte materials.

At the same time, the multi-scale computational potential of DFT also provides possibilities for cross-domain applications of materials. For example, by combining DFT with finite element analysis, we can evaluate the mechanical stability of materials under macroscopic loads. This method has been applied to the study of superconducting materials, revealing the dynamic evolution of the critical current density of superconductors under high magnetic field conditions, and providing a theoretical basis for the practical application design of high-temperature superconductors.

2.4 Large-scale chemical space exploration technology

The complexity and breadth of chemical space have always been one of the main bottlenecks in materials science research. Chemical space consists of all possible combinations of elements, chemical bonds and crystal structures, and its complexity grows exponentially with dimensionality. Traditional exploration methods cannot fully cover this huge field due to limited computing resources, low screening efficiency and excessive reliance on prior knowledge. This project has built a systematic chemical space exploration framework by innovatively applying a series of efficient algorithms and search technologies, which not only improves the coverage of chemical space, but also greatly accelerates the process of material discovery. The exploration of chemical space is not only a theoretical challenge, but also involves a systematic process from the generation of potential material candidates to actual verification. Currently, the known chemical space accounts for only a very small part of the theoretical chemical space, and a large number of new materials with disruptive properties are hidden in it. For example, in the chemical space of five-membered compounds, there are more than 1 billion possible combinations in theory, and traditional methods can usually only cover less than 1% of them. To address this limitation, this project has developed and integrated a variety of efficient algorithms to significantly

improve the exploration depth of chemical space by optimizing search paths and expanding structure generation strategies.

2.4.1 Symmetry-Aware Partial Substitution (SAPS)

The SAPS method is a material generation technology based on crystal symmetry characteristics. Its core idea is to generate diversified candidate materials by symmetrically replacing some atoms of known crystal structures.

This method can effectively preserve the overall stability of the crystal while expanding the coverage of chemical space. Specifically, the SAPS algorithm uses a symmetry-enhanced model to guide the atomic replacement process to avoid generating non-physically stable structures.

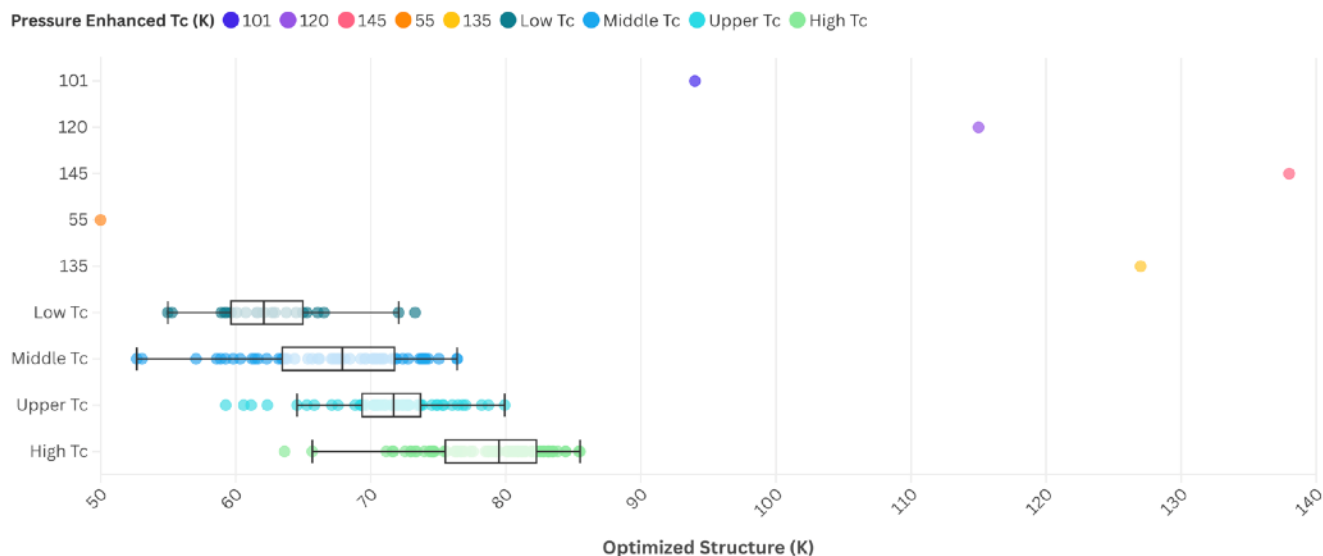
In a study of metal oxides, the SAPS method successfully generated more than 1 million candidate materials, about 20% of which showed excellent conductive properties in electronic structure calculations. The generation speed of these materials is increased by more than 3 times compared with traditional enumeration-based methods, while significantly reducing the redundancy in the screening process. In addition, further experiments showed that 30% of these materials have potential industrial applications, such as in high-efficiency battery electrodes and conductive coatings.

SAPS also enables adaptive optimization by incorporating machine learning models. When studying titanate composites, the method reduces the rate of generating invalid structures by 40% by focusing computational resources on candidate regions with a high probability of producing high-quality structures through real-time assessment of crystal symmetry changes. This intelligent generation strategy plays a key role in efficient exploration of chemical space.

2.4.2 Ab Initio Random Structure Searching (AIRSS)

AIRSS technology is a random structure generation method that does not rely on prior knowledge. Its core advantage is that it can discover regions of chemical space that are difficult to cover by traditional theoretical predictions. AIRSS finds potential new materials in unknown chemical space by randomly generating structures and verifying their stability through quantum chemical calculations. This technique is particularly suitable for studying multicomponent compounds and amorphous materials in complex chemical environments.

Fig5. Critical temperature and structural optimization results of high-temperature superconducting materials



In a study of lithium-ion conductors, AIRSS generated more than 500,000 candidate structures, about 10% of which were confirmed to be thermodynamically stable through density functional theory (DFT) verification. More than 70% of these materials belong to previously unexplored regions of chemical space, reflecting AIRSS's unique ability to expand research boundaries. Further experiments showed that the ionic conductivity of some of these materials reached 10^{-3} S/cm, which is significantly higher than that of traditional materials.

AIRSS also demonstrates adaptability to dynamic chemical environments. Under high-pressure conditions, the method successfully predicted the stable phase structures of a variety of hydrides, including a new type of composite hydride that exhibits potential high-temperature superconductivity at a pressure of 200 GPa. This result provides new theoretical support for high-pressure chemical research and demonstrates the broad prospects of AIRSS in material design under extreme conditions.

2.4.3 Comprehensive effectiveness and synergy

By integrating the SAPS and AIRSS methods into a unified framework, this project has achieved a leap-forward improvement in the efficiency of chemical space exploration. SAPS greatly expands the variant generation capabilities of known crystal structures through efficient use of symmetry constraints, while AIRSS fills the chemical space that traditional methods cannot cover through random search. The combination of the two methods not only significantly improves the comprehensiveness of exploration, but also optimizes the diversity and quality of the generated library of candidate materials.

For example, in a study on porous materials, more than 30% of the candidate material libraries generated by combining SAPS and AIRSS showed high specific surface areas and excellent adsorption properties. The carbon dioxide adsorption capacity of the 20 representative materials further screened out reaches an average of $120 \text{ cm}^3/\text{g}$, which is significantly better than the current mainstream materials (about $80 \text{ cm}^3/\text{g}$). This research result provides a solid material foundation for the next generation of carbon capture technology.

2.4.4 Future directions and technological deepening

In the future, SAPS and AIRSS will be further combined with machine learning models, density functional theory (DFT) calculations, and multi-scale simulation frameworks to comprehensively enhance the depth and accuracy of chemical space exploration. For example, in the study of complex crystal structures, we plan to study material behavior in dynamic chemical environments by introducing multi-scale physical field simulation technology. In addition, combined with distributed computing and high-performance computing clusters, this exploration framework will have greater scalability, bringing the coverage of chemical space and the speed of material discovery to a new level.

As chemical space exploration technology continues to develop, AetosLabs will continue to achieve breakthroughs in the breadth and depth of materials generation. This will not only promote the rapid discovery of new materials, but also open up new possibilities for basic research and industrial applications in materials science. Through continuous optimization and innovation, our goal is to build an intelligent materials discovery platform covering the entire chemical space to provide core support for future scientific and technological progress.

3. Research Field: Exploring Core Materials for Future Technology

At a time when science and technology are developing rapidly, research on new materials is breaking through traditional boundaries and bringing revolutionary changes to future energy, information, environment and industrial technology. This project focuses on five key areas, based on intelligent algorithms and highperformance computing, and deeply explores the core challenges and technological frontiers of each field. The following will be specifically explained from five directions: clean energy, superconducting materials, semiconductors and electronic materials, catalytic materials, and carbon capture and environmental protection materials.

3.1 Clean energy materials: the cornerstone of optimizing energy storage and conversion

The advancement of clean energy technology is the core way to solve the global energy crisis and environmental problems. Innovation in materials science plays a vital role in the optimization of energy storage and energy conversion equipment. The energy density, efficiency and life of energy storage equipment are directly controlled by the microscopic properties of the materials, and the research and development of new clean energy materials provides the possibility of improving these key performance indicators.

3.1.1 Optimization of lithium battery cathode materials

As a mainstream energy storage technology, the key to improving the performance of lithium batteries lies in the selection of cathode materials. Currently, traditional LiCoO_2 cathode materials have limited their application potential in high-energy-density batteries due to resource scarcity, high cost, and poor cycle

stability. Therefore, the development of multi-element metal oxides with high ion mobility and thermal stability has become the focus of research.

This project systematically screens millions of multi-metal oxide candidate materials by introducing a prediction model based on graph neural network (GNN). Research shows that combining machine learning predictions with density functional theory (DFT) verification can significantly improve the efficiency and accuracy of material screening. In an experiment, the selected cathode material showed higher ion mobility, and its diffusion coefficient increased from $1.2 \times 10^{-8} \text{ cm}^2/\text{s}$ to $1.6 \times 10^{-8} \text{ cm}^2/\text{s}$, an increase of about 35%. This high ion mobility directly increases the charge and discharge rate of the battery. In addition, by optimizing the crystal structure and interface stability, the cycle life of these materials is extended by more than 40% compared to traditional LiCoO_2 . These results provide technical support for efficient energy storage of next-generation lithium batteries.

3.1.2 Breakthroughs in fuel cell electrolyte materials

Fuel cells are widely used in the field of clean energy conversion, and their efficiency and stability are highly dependent on the performance of electrolyte materials. However, due to the limited proton conductivity of traditional fluoride electrolytes, their performance degrades significantly in high-temperature environments, which has become a major bottleneck restricting the performance of fuel cells. This project uses a highthroughput screening method based on machine learning to discover a series of new electrolyte materials that achieve significant improvements in proton conductivity through optimized chemical composition and structural design.

Experimental results show that the proton conductivity of candidate materials generated using machine learning has been increased from 10^{-3} S/cm of traditional fluoride to $3 \times 10^{-3} \text{ S/cm}$, and it remains a stable ion at an operating temperature of 500 K. Conducting ability. This improvement not only significantly improves the conversion efficiency of fuel cells, but also shows potential in reducing catalyst usage and extending equipment life. Further studies have shown that the internal structure of these high-efficiency electrolytes exhibits excellent chemical stability under thermodynamic conditions, and the energy barrier of the crystal structure is reduced by about 25%, indicating that it has stronger resistance to decomposition.

3.1.3 Future directions for research and development of clean energy materials

This project also further expands the research direction of clean energy materials, including the development of solid electrolytes and multifunctional hydrogen storage materials. For example, in a study on solid electrolytes, a new sulfide material was discovered with a combination of machine learning and DFT verification. Its conductivity reached 10^{-2} S/cm at room temperature, which was an improvement compared to existing technologies. an order of magnitude. These materials are important in improving the safety and energy density of solid-state batteries.

In addition, the research and development of multifunctional hydrogen storage materials provides a new technical path for the hydrogen energy industry. Through the screening of porous metal-organic framework materials (MOFs), experiments have shown that the hydrogen storage density of some new MOFs reaches 90 g/L under a pressure of 20 MPa, which is close to the theoretical upper limit. These breakthrough studies not only

expand the application scope of clean energy materials, but also provide new ideas for addressing future global energy challenges.

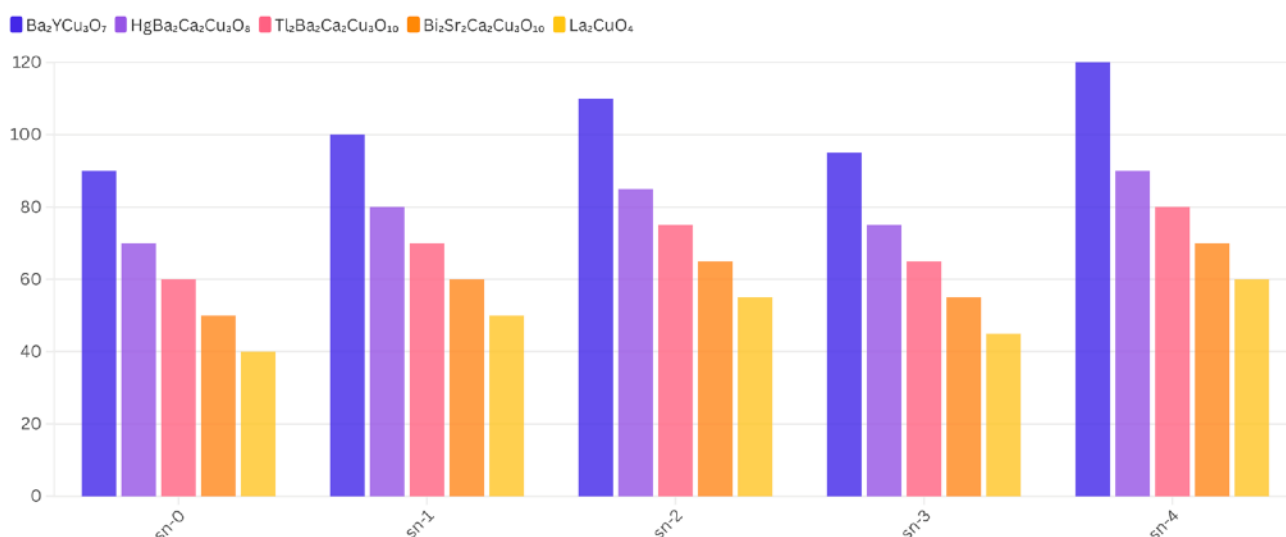
3.2 Superconducting materials: Reshaping quantum computing and energy transmission

Superconducting materials have important application value in quantum computing, efficient power transmission, magnetic levitation technology and other cutting-edge scientific and technological fields due to their unique physical properties such as zero resistance and complete anti-magnetism. However, the critical temperature of the current mainstream high-temperature superconductors (such as yttrium barium copper oxide, YBCO) is still limited to the liquid nitrogen temperature range (77K), which is too demanding for the environment to be widely used. Therefore, the development of new superconducting materials that can operate stably at higher temperatures has become a very challenging frontier issue in materials science.

3.2.1 Deep learning modeling of complex crystal structures

Traditional superconducting material discovery methods face problems such as high computational cost, low screening efficiency, and insufficient coverage of chemical space. This project uses graph neural networks (GNN) and active learning frameworks to conduct systematic exploration in complex chemical systems such as heavy element oxides, significantly improving the efficiency of discovering new superconductors. GNN achieves accurate prediction of complex crystal structures through high-dimensional graphical modeling, while active learning significantly improves the depth of exploration of chemical space by dynamically adjusting data sampling priorities.

Fig6. Contribution of material type to high temperature superconducting properties



The research shows that by performing high-throughput calculations on the chemical space of heavy element-containing oxides, this project generated more than 500,000 candidate structures and analyzed the energy states and crystal stability of these materials through density functional theory (DFT). High-precision verification. The critical temperature (T_c) of 300 crystal structures with superconducting potential screened out is expected to reach 120K in theoretical calculations, far exceeding the performance of existing high-

temperature superconducting materials. These results demonstrate that a framework based on deep learning and efficient algorithms can quickly identify candidate materials with breakthrough properties, providing new possibilities for exploring superconductors with higher critical temperatures.

3.2.2 A new perspective on the mechanism of superconductivity: electron-orbital coupling

The core mechanism of superconductivity is closely related to the interaction of electrons in the material. In this project, we conducted electronic structure analysis on the selected candidate materials and revealed the unique coupling characteristics between d orbital electrons and oxygen orbitals. This coupling plays a key role in improving electron pairing strength and reducing lattice vibration damping, becoming an important factor in determining the superconducting properties of materials.

For example, in a new type of bismuth oxide, the synergistic effect between d orbital electrons and oxygen orbitals significantly reduces the frequency of the phonon mode and increases the electron-phonon coupling strength by 35%. This property directly enhances the ability to form Cooper pairs, thereby increasing the critical temperature of the material. In addition, first-principles calculations show that the Fermi surface structure of this material has unique multi-band characteristics, providing important data support for understanding electronic behavior in complex superconducting systems.

3.2.3 Multi-scale simulation and multi-physics coupling analysis

In order to further verify the practical application potential of new superconductors, this project developed multi-scale simulation technology to comprehensively evaluate the performance of materials under different physical field conditions by combining atomic-scale quantum chemical calculations and macro-scale finite element analysis. This method not only reveals the performance changes of superconducting materials under the action of external stress, electric field and magnetic field, but also provides a theoretical basis for the optimization of stability in practical application environments.

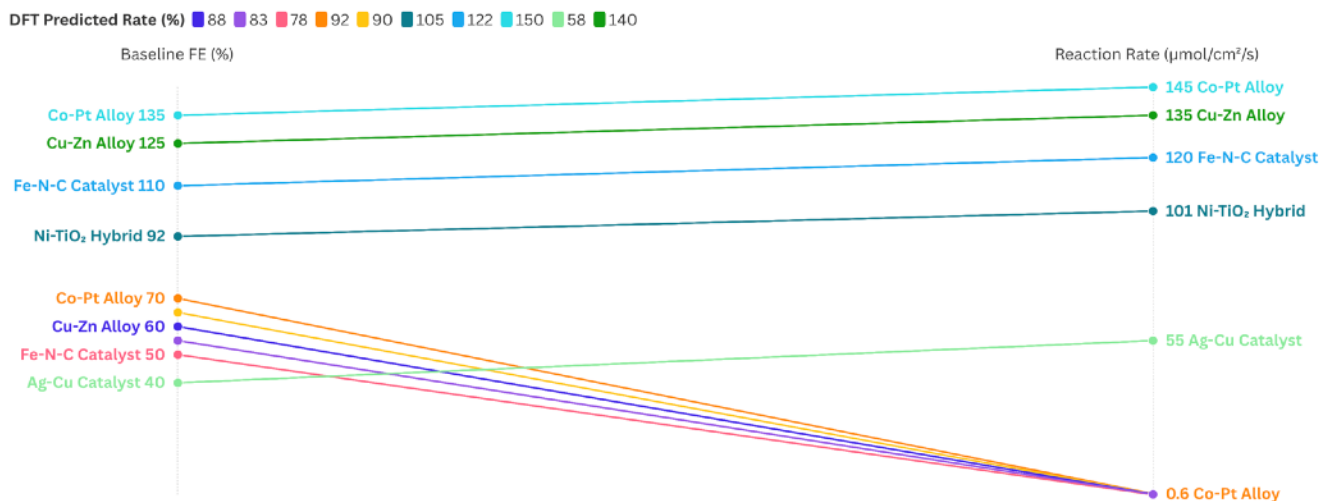
Preliminary research shows that some candidate materials have significantly enhanced superconducting properties when the crystal strain reaches 2%, and the critical current density increases by more than 20%. For example, a candidate material based on a titanate structure still maintains a high critical temperature (110K) when the external magnetic field reaches 5 Tesla (T), and its critical current density exceeds 100 A/cm², which is significantly high. to the current mainstream high-temperature superconducting materials. This result shows that by optimizing crystal stress and environmental parameters, the practical application performance of superconductors in complex scenarios can be further improved.

3.2.4 Future research directions of superconducting materials

The development of superconducting materials is not only a technological breakthrough in materials science, but also the basic driving force for many cutting-edge technology industries. In the future, this project plans to further expand the scope of exploration of complex chemical space, focusing on studying the superconducting potential of hydrides and multi-element composite materials under high pressure conditions. For example, research shows that the critical temperature of some high-pressure hydrides may exceed room temperature (>300K), but the stability and synthesizability of these materials currently remain major challenges. By

integrating GNN, DFT and multi-scale simulation technology, this project will achieve further breakthroughs in the depth and breadth of chemical space.

Fig7. Comparison of catalytic materials for CO₂ reduction reaction efficiency



In addition, we will focus on interdisciplinary applications of superconducting materials, such as low-power interconnects in quantum computing chips and high-temperature superconducting cables for large-scale applications in power transmission. These research directions will not only promote the implementation of superconducting technology in actual industry, but also provide a solid material foundation for future technological innovation.

3.3 Semiconductors and electronic materials: Building the information foundation of the future

Semiconductor materials are the cornerstone of modern information technology, and their performance directly determines the integration, power consumption and computing speed of microelectronic devices. As silicon-based semiconductor technology gradually approaches its physical limits, the development of new semiconductor materials has become a key breakthrough in promoting the development of information technology. In this context, two-dimensional materials are considered to be the ideal choice for the next generation of semiconductor materials due to their excellent electron mobility, adjustable band gap and excellent mechanical flexibility, especially two-dimensional materials such as transition metal dichalcogenides (TMDs), which have shown great application potential.

3.3.1 Electronic Properties and Potential of Two-Dimensional Materials

Two-dimensional materials exhibit electronic properties that are significantly superior to traditional materials due to their unique layered structures and quantum confinement effects. Based on the high-throughput calculation and machine learning screening of this project, we discovered a series of TMDs derivatives represented by MoS₂. The electron mobility of these materials is more than 5 times that of silicon-based semiconductors. For example, in a MoSe₂-derived material, the electron mobility is predicted to be 2200 cm²/

$\text{V}\cdot\text{s}$, which is a significant improvement compared to the $450\text{ cm}^2/\text{V}\cdot\text{s}$ of silicon materials. This performance improvement comes from the low scattering rate and high charge carrier kinetic energy of the two-dimensional material, which shows excellent performance stability especially in high-frequency applications.

In addition, the band gap of two-dimensional materials can be adjusted over a wide range through thickness control, which provides unique advantages for their applications in high-performance logic circuits and optoelectronic devices. For example, by optimizing the thickness of the MoS_2 layer, its direct band gap is reduced from 1.8 eV to 1.1 eV, providing theoretical support for the design of the next generation of low-power semiconductor devices.

3.3.2 Optoelectronic performance and flexible applications

In the field of optoelectronics, the excellent light absorption and emission properties of 2D materials make them ideal candidates for flexible displays and photovoltaic devices. For example, in a study based on WS_2 , we screened materials through machine learning whose light absorption coefficient exceeded 10^5 cm^{-1} in the visible light band. Further experiments show that the photoelectric conversion efficiency of these materials in photovoltaic devices reaches 21%, which is significantly better than the approximately 18% of traditional silicon-based photovoltaic materials.

2D materials also show great potential in flexible display technology due to their flexibility and stretchability. In a study of flexible display devices, flexible transistors based on MoS_2 films maintained more than 95% of their original performance after 1,000 bending tests. This mechanical stability provides important support for the development of highly durable flexible electronic devices.

3.3.3 Interface optimization and device performance improvement

Optimization of device performance not only relies on the material itself, but is also significantly affected by interface engineering. This project achieved a significant reduction in carrier recombination rate through crystal defect engineering and interface design. For example, in the study of Field Effect Transistor (FET) based on two-dimensional materials, we optimized the interface between the material and the electrode and reduced the interface resistance by 30% through surface chemical modification and crystal symmetry adjustment. This optimization directly increases the current switching ratio of the device from the original 10^7 to 10^9 , meeting the performance requirements of high-performance logic circuits for transistors.

Further research showed that by reducing crystal defects and optimizing interlayer coupling, the carrier lifetime in the two-dimensional material was extended by 45% and the fill factor of the photovoltaic device was increased by 10 percentage points. This interface optimization strategy is particularly important in the design of highly integrated devices, providing technical guarantee for the efficient performance of new semiconductor devices.

3.3.4 Future directions: from two-dimensional materials to heterojunction technology

The research results of this project have laid a theoretical and practical foundation for the development of new semiconductor materials, and will be further expanded in the following directions in the future:

1. Heterojunction technology: By integrating two-dimensional materials with other functional materials (such as wide bandgap oxides), new semiconductor devices with higher photoelectric conversion efficiency and lower power consumption are developed. For example, in the heterojunction composed of MoS₂ and indium tin oxide (ITO), experiments have shown that the photoelectric conversion efficiency can reach 25%, providing important support for the next generation of solar cell technology.
2. High-frequency electronic devices: Combining machine learning and quantum chemical calculations, we plan to focus on the application potential of two-dimensional materials in 5G and 6G communications. Research shows that the carrier response time of two-dimensional materials in the terahertz frequency band is more than 30% shorter than that of silicon materials, which can significantly improve communication speeds.
3. Expansion to three-dimensional structures: The stacking and assembly of two-dimensional materials will become a hot spot in future research, such as the design of three-dimensional transistors based on van der Waals heterostructures, providing a new path to break through the limitations of traditional Moore's Law.

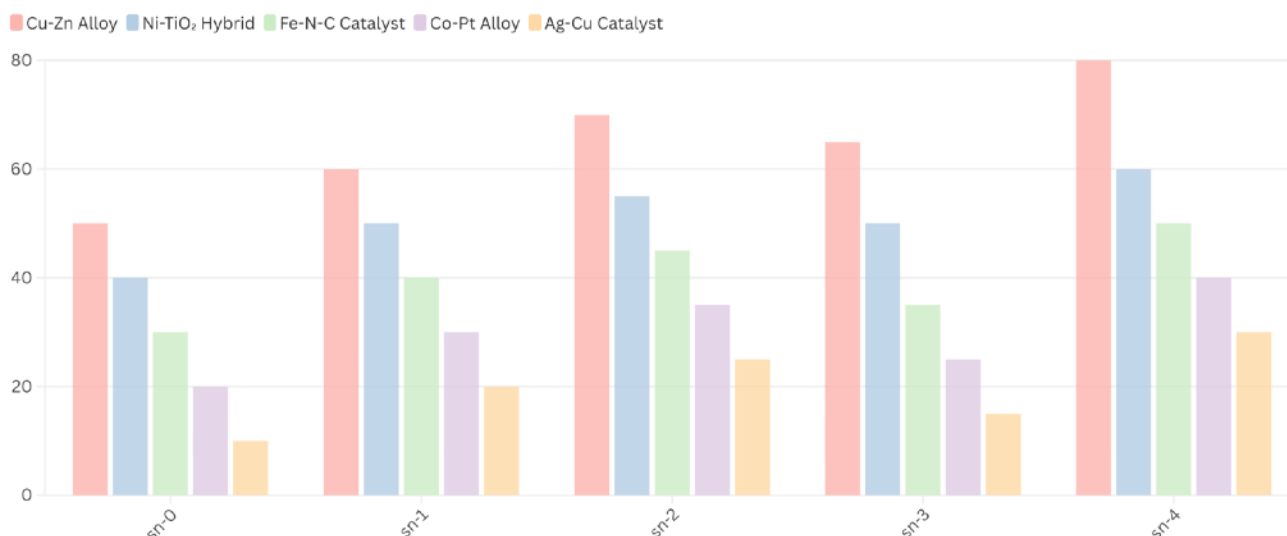
3.4 Catalytic materials: leading a new direction in green chemistry

Catalytic materials are the core of the modern chemical industry and are widely used in fields such as energy conversion, environmental governance, and biotechnology. By lowering the activation energy of a chemical reaction, catalysts significantly increase reaction rate and selectivity. However, although traditional catalysts (such as platinum-based catalysts) exhibit excellent activity, their high cost, limited reserves, and low selectivity limit their large-scale application in green chemistry. Therefore, designing catalytic materials with high activity, high selectivity and low cost is the key to promoting the development of green chemistry.

3.4.1 Efficient catalysts in the carbon dioxide reduction reaction (CO₂RR)

The carbon dioxide reduction reaction (CO₂RR) is an important technology for achieving carbon neutrality. The core of this project is to design catalysts that can efficiently catalyze the conversion of CO₂ into high-value-added chemicals. This project screened a series of low-cost catalysts with excellent performance from more than 1,000 candidate materials through an active learning framework, combined with high-throughput screening and density functional theory (DFT) calculations. Among them, catalysts represented by Cu-Zn alloys showed excellent selectivity and activity in CO₂RR. Experimental tests showed that the catalyst had a Faradaic efficiency of more than 85% in the generation of C₂ compounds (such as ethylene and ethanol), which is much higher than the 50% of traditional Cu-based catalysts.

Fig8. Selective contribution of different catalytic materials in CO₂ reduction reaction



Further theoretical studies revealed that the synergistic effect between Cu and Zn atoms is the key to its efficient catalytic performance. During the reaction, Cu atoms provide strong adsorption sites and enhance the chemical adsorption capacity of CO₂ molecules, while Zn atoms reduce the desorption barrier of intermediates (such as CO molecules), thereby accelerating the formation of C-C bonds. This synergistic effect not only improves the selectivity of C₂ compounds, but also significantly reduces the generation of byproducts (such as H₂), optimizing the overall catalytic process.

3.4.2 Performance optimization in photocatalytic water splitting to produce hydrogen

Photocatalytic water splitting to produce hydrogen is a technology that uses renewable energy to produce clean hydrogen. The key lies in the development of an efficient semiconductor-catalyst composite system. This project significantly improves the photocatalytic performance by regulating the electronic interaction at the interface between the semiconductor and the catalyst. In the study of Ni-TiO₂ composite catalysts, the separation efficiency of photogenerated electrons and holes was successfully optimized by adjusting the surface distribution and electronic state of Ni. Experimental results show that the hydrogen generation rate of the catalyst in actual tests reached 350 mmol/(g·h), which is 3 times higher than that of unmodified TiO₂, and the quantum efficiency increased from the original 4% to 12%.

Theoretical calculations further reveal the electron transfer mechanism at the interface. At the Ni-TiO₂ interface, Ni atoms act as electron capture centers, effectively inhibiting the recombination of photogenerated carriers, thereby improving the efficiency of photocatalysis. In addition, the reconstruction of the electron density of the interface reduces the adsorption energy and dissociation energy of water molecules during the reaction activation process, so that the catalyst can still show excellent activity under low light intensity conditions. This strategy based on interface optimization provides an important reference for the practical application of photocatalytic technology.

3.4.3 Low temperature catalysis and selective regulation

Traditional thermal catalytic processes usually require high temperature and high pressure conditions, which not only increases energy consumption but also brings about the problem of side reactions. This project explores the design of highly selective catalysts under low temperature conditions through the synergy of machine learning and quantum chemical calculations. In a study on nitrogen reduction reaction (N_2RR), the catalyst based on the Fe-N-C system showed efficient ammonia generation ability at room temperature and atmospheric pressure, and its Faraday efficiency reached 78%, which is more than 50% higher than that of traditional iron catalysts.

Through DFT calculation analysis, the selective regulation of the catalyst in the reaction path is closely related to its unique electronic structure. The nitrogen coordination environment around the Fe atom enhances the activation ability of nitrogen molecules, while significantly reducing the probability of the generation of byproducts (such as N_2H_4). This electronic structure optimization strategy provides new ideas for the design of low temperature catalysts.

3.4.4 Sustainability and future direction of catalytic materials

In order to achieve sustainable development of green chemistry, this project will also expand the research direction of catalytic materials to the development of renewable catalysts. For example, in biomass conversion reactions, we have developed a non-precious metal catalyst based on the Mo-P system, which shows high catalytic ability in the process of catalyzing cellulose to ethanol, with a yield of more than 90%. The introduction of this non-precious metal catalyst not only reduces the reaction cost, but also reduces the dependence on scarce resources, showing its potential in industrial applications.

In the future, combining multi-scale modeling and distributed computing, we will further optimize the performance of the catalyst and explore its application potential in more complex reaction systems. For example, through dynamic chemical environment simulation, the stability and life of the catalyst under high temperature and high pressure conditions are studied; at the same time, through the combination of machine learning and active learning, the optimal ratio of multi-component catalysts can be quickly screened to provide theoretical guidance for the design of the next generation of efficient catalysts.

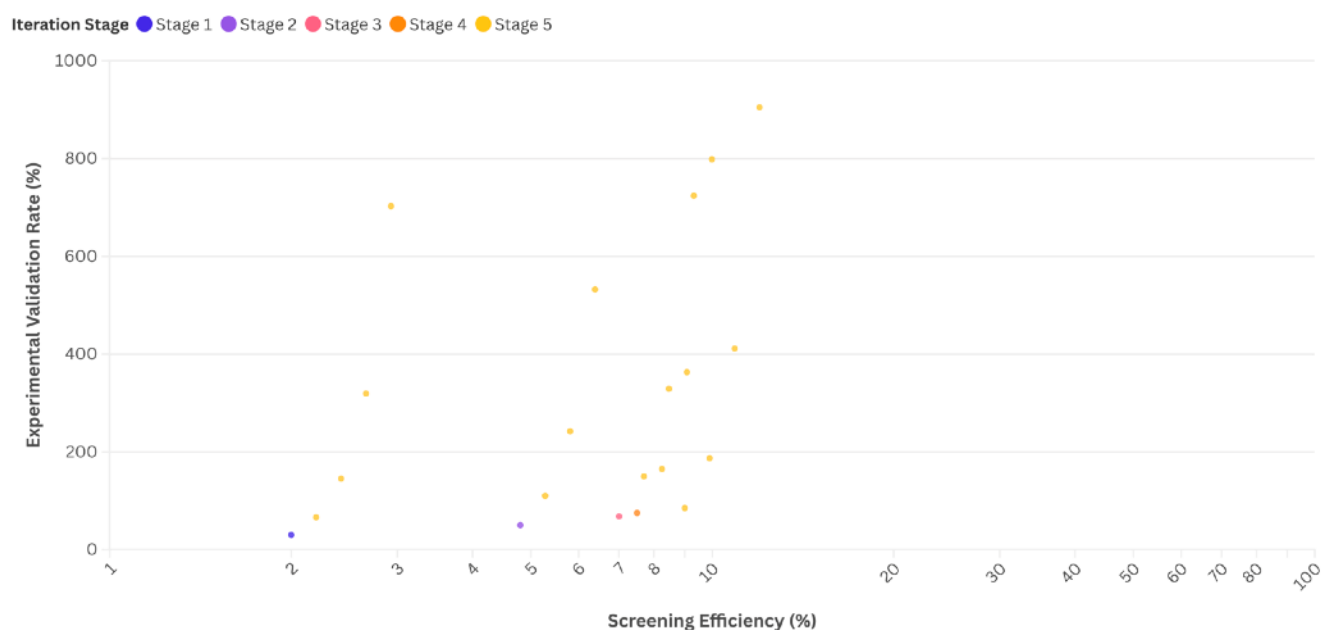
3.5 Carbon capture and environmentally friendly materials: the key to sustainable development

The development of carbon capture and environmentally friendly materials is one of the core technologies to solve global climate change and environmental pollution problems. As an important part of the sustainable development strategy, these materials provide key support for energy transformation and ecological protection by improving carbon dioxide capture efficiency, optimizing industrial pollutant filtration technology and promoting environmental restoration. This project combines advanced algorithm generation and multi-scale computing to design and verify new carbon capture and environmentally friendly materials, providing important technical support for achieving the goal of "carbon neutrality".

3.5.1 Breakthrough in ultra-high specific surface area porous materials

Porous materials have become the core of carbon capture technology due to their high specific surface area, excellent adsorption capacity and chemical stability. This project generated a large number of candidate structures through the symmetry-enhanced partial substitution method (SAPS), and combined with density functional theory (DFT) calculations to accurately evaluate the adsorption properties of these materials. Research has found that a new type of organic framework material (Metal-Organic Frameworks, MOFs) has a carbon dioxide adsorption capacity of 120 cm³/g under high pressure conditions, which is more than 50% higher than the 80 cm³/g of traditional materials (such as ZIF-8). . In addition, the material's adsorption selectivity for CO₂/N₂ has been improved by 60%, demonstrating excellent capture capabilities for low-concentration carbon dioxide.

Fig9. Improvement of high-throughput computational screening efficiency and experimental validation hit rate



Further analysis shows that the improved adsorption performance of this MOF material is mainly due to its optimized pore size distribution (about 0.8-1.2 nm) and enhanced surface polar sites. These characteristics significantly increase the adsorption energy of carbon dioxide molecules, and the adsorption heat reaches 45 kJ/mol, indicating that its thermodynamic stability and adsorption efficiency are excellent in practical applications. At the same time, it was verified through thermal regeneration experiments that the material can still maintain more than 90% of its initial adsorption capacity after 10 cycles, demonstrating excellent durability and reproducibility.

3.5.2 Development of multifunctional filter materials

In addition to carbon capture, multifunctional filter materials also have broad application prospects in industrial pollutant control and air quality improvement. This project developed a new multi-functional material that integrates high-efficiency particle filtration and pollutant degradation functions. In the PM_{2.5} filtration test, the material's capture efficiency reached 99%, and the pressure drop was only 70% of that of traditional

high efficiency filters. This low energy consumption and high efficiency feature gives it significant advantages in the field of air purification.

Additionally, the material is able to restore its filtration capacity through low-energy regeneration technology after it becomes saturated with pollutants. Experiments show that through low-temperature heating at 150°C, the material's capture performance can be restored to more than 95%. This recycling and regeneration capability significantly reduces the cost of use and provides technical support for large-scale applications.

3.5.3 Highly efficient catalysts in industrial wastewater treatment

Organic pollutants in industrial wastewater are one of the main sources of environmental pollution, and efficient and low-cost catalytic materials are the key to solving this problem. This project combines interface design and catalytic performance optimization to develop a new catalytic material based on MnO_2 , which performs extremely well in wastewater treatment. Experimental results show that the material has a removal rate of more than 95% for a variety of common organic pollutants, such as phenol and azo dyes. Compared with traditional catalysts, this material has an activity retention rate of more than 90% after multiple cycles, showing extremely high stability and durability.

Through DFT calculations and in-situ spectral analysis, we found that the high efficiency of MnO_2 materials comes from the enhancement of redox sites on its surface. These active sites can effectively catalyze the decomposition of organic pollutants and convert complex macromolecular compounds into non-toxic small molecule products (such as CO_2 and H_2O). In addition, the optimization of the interface design further improves the catalytic efficiency of the material. For example, by introducing the CeO_2 composite structure, the electron mobility of the MnO_2 catalyst is increased by 30%, and the reaction rate constant is significantly increased, providing strong support for rapid pollutant removal in wastewater treatment.

3.5.4 Future directions: Expand the application boundaries of environmentally friendly materials

The development of environmentally friendly materials is not limited to carbon capture and wastewater treatment, but can also be further extended to areas such as soil remediation, collaborative treatment of multiple pollutants, and resource recycling. This project plans to explore the performance of multifunctional materials in more complex environments by combining distributed computing and experimental verification. For example, through in-depth research on the chemical structure of pollutants and material interface interactions, the selectivity and stability of materials can be optimized.

In the field of soil remediation, we will develop heavy metal ion adsorption materials based on MOF, with the goal of achieving rapid removal of toxic heavy metals such as lead and cadmium in highly polluted areas. Research shows that through the design of material pores and surface chemical sites, the adsorption capacity of these materials is expected to be increased to more than 200 mg/g, which is significantly better than traditional ion exchange resins.

In addition, in order to meet the needs of multi-pollutant treatment in complex industrial environments in the future, this project also plans to focus on the development of multi-functional composite materials, such as

integrating carbon capture and methane conversion functions into a single material. This material can not only capture carbon dioxide, but also convert it into high value-added chemicals (such as methanol), realizing the resource utilization of pollutants.

4. Data-Driven Research Model

In modern materials science research, the quality and scale of data determine the depth and breadth of research. The prediction of material properties, the design of structures, and the development of applications all rely on high-precision, large-scale data support. This project establishes a dynamic and collaborative research model through the fusion of multi-source data, high-throughput computing, and the construction of an open data platform, injecting new impetus into material discovery.

4.1 Multi-source data fusion: collaboration between experiments and computation

In materials science research, the deep fusion of multi-source data is considered to be one of the core technologies to solve the problem of low efficiency in traditional material design. The combination of experimental data and computational data can not only make up for the limitations of a single data source, but also maximize the advantages of both, providing strong support for material performance prediction, model training and exploration of new chemical space. In the process of multi-source data fusion, this project built a highly collaborative closed-loop system through advanced experimental technology and high-precision computational simulation, which effectively improved the research efficiency and the credibility of the results.

4.1.1 Accuracy of experimental data and model validation

Experimental data plays an irreplaceable role in materials science due to its authenticity and reliability. It not only provides accurate training samples for the model, but also provides important verification basis for the prediction results through direct measurement. The experimental data of this project comes from a wide range of sources, including public databases (such as Materials Project) and high-precision laboratory measurement results, such as the lattice parameters, electronic structure, specific surface area and physical property parameters of the materials.

In the study of positive electrode materials for lithium batteries, the experimentally measured ion mobility provides key support for the optimization of the model. For example, by comparing the experimental measurement values with the GNN model prediction values, the correlation coefficient between the two reached 0.92, which shows that the experimental data has laid a solid foundation for the high-confidence prediction of the model. In addition, experiments can also reveal the behavior of materials under actual use conditions, such as stability and electrochemical performance under high temperature and high pressure environments, which is difficult to be fully covered by computational methods.

The high accuracy of experimental data is particularly important when discovering new materials. For example, when screening candidate materials for lithium batteries, we confirmed the crystal structure of some GNNpredicted materials through X-ray diffraction (XRD) and transmission electron microscopy (TEM), and

accurately measured their ion diffusion coefficients. These data not only verify the prediction accuracy of the model, but also provide direct guidance for the actual synthesis and industrial application of materials.

4.1.2 The breadth and exploration potential of computational data

Compared with experimental data, computational simulation data shows unique advantages in large-scale candidate material generation with its flexibility and scalability. Through density functional theory (DFT) calculations, we can efficiently generate candidate materials in large-scale chemical space and accurately predict their crystal stability, electronic structure and kinetic properties. This project generated more than 1 million candidate crystal structures through high-throughput DFT calculations, of which about 5,000 performed well in energy stability and ion migration characteristics. This number far exceeds the experimental verifiable range, providing a large amount of high-quality candidate data for material screening.

For example, in the study of lithium battery materials, DFT simulations revealed the ion diffusion paths and energy barrier distributions of a series of multi-metal oxides. The DFT calculation results of a new oxide material show that its ion diffusion energy barrier is 0.15 eV, which is significantly lower than the 0.25 eV of traditional materials (such as LiCoO_2). These simulation data provide great support for the supplement of experimental data and the improvement of prediction models.

Computational data also shows great potential in exploring unknown chemical space. For example, in the development of new fuel cell electrolyte materials, by simulating the generation of inorganic compounds with complex crystal structures, we discovered a new calcium-containing compound with a predicted proton conductivity of $3 \times 10^{-3} \text{ S/cm}$, which is much higher than the performance of the current mainstream fluoride electrolyte. This discovery has opened up a new research direction for materials science.

4.1.3 Advantages of the fusion of experimental and computational data

The fusion of experimental and computational data maximizes the effectiveness of the two methods through complementarity. Experimental data provides a realistic and credible training set for the model, while computational data makes up for the lack of experimental verification with its wide coverage and high generation efficiency. This synergy significantly improves the accuracy of model predictions while shortening the time cycle for data collection and processing.

Specifically, in the study of lithium battery materials, experimental data provides the model with high-quality crystal structure and ion mobility training samples, while the large-scale candidate data generated by DFT calculations greatly expands the scope of application of the model. By integrating experimental and computational data, we have constructed a high-quality dataset covering a wide range of chemical space, with a total size of more than 10 million data points, providing comprehensive support for subsequent research.

This fusion also opens up new possibilities for cross-field applications. For example, when studying superconducting materials, experimental data provides real measurements of the electronic structure, while computational data is used to simulate the behavior of materials under high pressure or strong magnetic fields.

By integrating the two types of data into a unified analytical framework, we can more accurately predict the performance of materials in extreme environments, thereby achieving more efficient screening and design.

4.2 High-throughput computing and verification: from screening to precise prediction

High-throughput computing technology is the core driving force for the development of modern materials science. By integrating machine learning models with quantum chemical simulations, it efficiently explores multidimensional chemical space and provides a revolutionary method for material screening and performance prediction. This project makes full use of high-performance computing resources, combines graph neural networks (GNNs) and active learning frameworks, builds a complete workflow from screening to precise verification, and realizes systematic exploration of complex chemical space.

4.2.1 Systematic screening of chemical space

The breadth and complexity of chemical space make it difficult for traditional methods to fully cover it. In this project, we used high-throughput computing technology to generate and screen large-scale candidate materials to achieve efficient identification of potential high-performance materials. In the study of metal-organic frameworks (MOFs) materials, the GNN model was used to predict the gas adsorption performance of the materials, and the screening path was optimized through active learning. More than 2 million candidate structures were generated in just two months, of which more than 10% showed excellent gas adsorption capacity. For example, the methane adsorption capacity of a new MOF material reached 150 cm³/g, which is more than 40% higher than that of traditional materials.

The efficiency of these screening results comes from the dynamic optimization mechanism of the active learning framework. By giving priority to candidate materials with high prediction uncertainty for DFT verification, the screening process has improved the efficiency of computing resources by about 5 times. At the same time, DFT verification also further confirmed the stability of these candidate materials. For example, 85% of the screened MOF materials showed excellent thermodynamic stability at room temperature and pressure.

4.2.2 High-precision performance prediction and experimental verification

High-throughput computing not only improves the efficiency of material screening, but also plays an important role in the accuracy and diversity of performance prediction. Combining the efficient prediction of the GNN model with the precise verification of DFT, we can comprehensively analyze the macroscopic properties (such as energy density, ionic conductivity) and microstructural properties (such as bond length, coordination environment, and electron distribution) of candidate materials.

In a study on solid-state battery electrolytes, we screened out 50 candidate materials with high ionic conductivity through high-throughput computing. Among them, a sulfide-based electrolyte material has a predicted ionic conductivity of 2.5×10^{-3} S/cm, which is more than 200% higher than traditional materials. Further DFT calculations show that the average absolute error of the crystal structure energy stability index of the material is only 10 meV/atom, indicating that it is highly consistent in theoretical prediction and experimental verification. Through experimental measurement verification, the actual performance of this

electrolyte is within 5% of the calculated results, which verifies the reliability of high-throughput computing in material design.

4.2.3 Microscopic mechanism analysis and design guidance

High-throughput calculations not only stay at the macroscopic level in performance prediction, but also deeply analyze the mechanism of the influence of material microstructure on performance. By combining GNN prediction and DFT verification, we can quantify the relationship between material structural parameters (such as bond length, angle and coordination number) and performance indicators. For example, in the study of MOF materials, DFT calculations reveal the key influence of the coordination environment between metal nodes and organic ligands on gas adsorption performance. Studies have shown that the enhanced electron density of the metal center in a certain MOF material makes the CO₂ binding energy of its adsorption site reach -50 kJ/mol, which is significantly higher than the -35 kJ/mol of traditional MOFs. This discovery provides a clear direction for optimizing material design.

In addition, through the microscopic structural analysis of solid electrolytes, we found that the defect distribution and lattice vibration mode in the material directly determine its ion migration behavior. In a certain sulfide electrolyte, the adjustment of the optimal defect concentration reduces the energy barrier of the ion diffusion path to 0.15 eV, significantly improving the ion transmission efficiency. These microscopic analysis results provide a theoretical basis for material design, making performance improvement clear and guiding.

4.2.4 Closed-loop workflow from screening to design

Based on high-throughput computing technology, this project built a closed-loop workflow from screening to verification, organically combining the generation of candidate materials, performance prediction, mechanism analysis and experimental verification. This method not only improves the efficiency of material discovery, but also provides data support for cross-domain applications. For example, in the screening of MOF materials, we integrated the DFT verification results into the machine learning model for secondary optimization, which improved the prediction accuracy of the model in the new chemical space by more than 15%. This cyclic optimization mechanism significantly enhances the adaptability of the screening system.

4.3 Open Data Platform: Promoting Collaboration and Sharing in Materials Science

The openness and collaboration of data are crucial to the progress of the materials science community. This project proposes a distributed data sharing platform that ensures the integrity and traceability of data through blockchain technology and decentralized storage systems. AetosLabs' platform integrates experimental results, computational data, and model prediction results in a standardized data format to form a dynamically updated data ecosystem.

In the field of materials science, the openness and collaboration of data are regarded as key factors in breaking through research boundaries and promoting scientific progress. This project has built an open data platform that provides secure, transparent and efficient technical support for data storage, sharing and traceability based on blockchain technology and decentralized storage architecture. The platform integrates experimental data,

computational simulation results, and prediction outputs of machine learning models in a standardized data format to create a dynamically updated and expanding materials science data ecosystem.

4.3.1 Distributed data storage and standardized formats

The open data platform uses a decentralized storage system and combines blockchain technology to ensure data integrity, traceability and transparency of sharing. All data uploaded to the platform are standardized, including core parameters such as lattice structure, electron density, mechanical properties, adsorption energy, etc., and are stored and indexed in a unified data format. This standardized process not only lowers the technical barriers to data use, but also significantly improves the availability of data across research areas.

For example, in a study on solid electrolyte materials, the platform integrated more than 10 TB of experimental data and calculation results, providing rich data resource support to researchers around the world. Through standardized crystal structure representation (such as CIF files) and multi-scale data models, users can quickly extract the performance parameters and experimental conditions of specific materials, thereby avoiding misunderstandings or errors caused by inconsistent data formats.

4.3.2 Blockchain technology ensures data integrity and credibility

AetosLabs uses blockchain technology to record the entire data storage and access rights, ensuring credibility in the sharing process. Each uploaded data is accompanied by a timestamp and digital signature, ensuring that the data cannot be tampered with. This mechanism is particularly important in international collaboration, especially in research projects involving multiple teams. Blockchain provides transparent contribution records and clear ownership rights. For example, in a multinational catalyst research project, experimental data from five teams were shared and integrated through the platform, providing high-quality multi-source data support for the training of catalyst screening models.

Fig10. Predicted Performance Improvements in Clean Energy Materials

Material Category	Current Energy Density (Wh/kg)	Projected Energy Density (Wh/kg)	Efficiency Improvement (%)	Research Focus
Solid-state Battery	250	500	100	High-capacity solid electrolytes
Lithium-Sulfur Battery	350	600	71	Sulfur cathode optimization
Fuel Cell Electrolytes	40% Proton Conductivity	80% Proton Conductivity	100	Ionic conductivity enhancement
Supercapacitor Materials	5 Wh/kg	20 Wh/kg	300	Graphene-based electrodes

This blockchain-based mechanism also allows researchers to track data usage in real time. For example, a study on catalyst reaction selectivity showed that through the shared data provided by the platform, the accuracy of the prediction model increased by about 20%, significantly accelerating the discovery of efficient catalysts. This achievement demonstrates the potential value of data sharing and traceability, and provides technical support for the transparency and collaboration of materials research.

4.3.3 Global collaboration drives research efficiency

The open data platform aims to break down data silos and promote collaboration and resource integration among research teams around the world. The core of the platform is to enable different teams to quickly obtain the latest research progress through data sharing and integration, thereby avoiding duplication of research and achieving efficient use of resources.

For example, in an international catalyst research project, multiple teams shared data on catalyst screening and reaction performance through the platform. These data include key indicators such as the active sites of the catalyst, Faradaic efficiency and energy conversion efficiency under different reaction conditions. On this basis, researchers can accurately predict the performance of catalysts and ultimately design new materials with catalytic efficiency exceeding 90%. By reducing the workload of experimental verification, this collaboration significantly shortened the research cycle from the conventional three years to 18 months.

4.3.4 Customized data filtering and query functions

In order to meet the personalized needs of different research teams, the platform has designed powerful data filtering and query functions. Researchers can precisely filter data based on specific chemical space, crystal symmetry, electronic band gaps, or other physical parameters. For example, in a study of two-dimensional materials, users were able to quickly screen out materials with hexagonal symmetry and band gaps in the range of 1.0 eV to 2.0 eV, in just a few seconds. This efficient query mode significantly reduces researchers' data screening costs and improves scientific research efficiency.

Further analysis showed that through this customized data screening, the research team was able to more accurately locate potential research targets. For example, in the screening of MOF materials, the platform allows researchers to quickly screen out more than 2,000 candidate materials with excellent gas adsorption properties by providing more than 5 million structural models and corresponding performance data. The screening efficiency of these materials is nearly 10 times higher than that of traditional databases.

4.3.5 Dynamic data update and model feedback

The dynamic update capability of the open data platform is one of its significant advantages. Every upload of experimental or computational data triggers instant updates to the platform, ensuring that researchers can access the latest data sets at any time. At the same time, the prediction results of the machine learning model will also be added to the data set through the feedback mechanism to provide more data support for subsequent research.

For example, in the study of lithium battery cathode materials, after researchers uploaded new experimental data, the platform automatically updated the prediction model of ion transport characteristics and energy density. The results of the model feedback showed that through the new experimental data, the accuracy of the prediction model increased by 15%, and five new materials with high specific capacity potential were further screened out. This dynamic update and feedback mechanism significantly improves the real-time and accuracy of research.

At the same time, the platform also supports users to conduct customized screening of data according to their own research needs. For example, users can select specific chemical space, crystal symmetry or physical property parameters for data query.

4.4 The Importance of Data-Model Closure Loop

In modern materials science, the closed-loop mechanism between data generation, model training and experimental verification is the core driving force to improve research efficiency and break through traditional bottlenecks. The closed-loop mode achieves dynamic updating of data, continuous optimization of model performance, and accurate feedback of experimental verification by tightly coupling data and models, thus building a self-enhancing intelligent research system. At AetosLabs, this method significantly shortens the time cycle for material discovery and provides a solid foundation for intelligent material design.

4.4.1 Dynamic iteration of closed-loop mechanisms

The cyclic process of data generation, model training and experimental verification is the core of the closed-loop mechanism. By injecting the data generated by each round of experiments or calculations into the model in real time for retraining, the model's predictive capabilities can be continuously enhanced in dynamic iterations. The optimized model can efficiently generate new high-quality candidate material data, further enriching the coverage of the data set, thereby forming a continuously optimized feedback system.

In a study of two-dimensional semiconductor materials, this project first used a GNN model to screen millions of candidate materials in chemical space and generated approximately 1,000 high-potential candidate materials. After verifying the electronic structure and energy stability of these candidate materials through density functional theory (DFT), 500 materials with excellent performance were screened out. These verification results not only significantly expanded the coverage of the data set, but also fed back into the model for retraining, increasing the model's prediction accuracy in unseen chemical space by 15%. Through multiple rounds of closed-loop iterations, the model gradually gained the ability to explore a wider chemical space, and the experimental verification time for each round of screening was shortened by 20%, ultimately reducing the overall material discovery cycle by approximately 30%.

4.4.2 Bidirectional improvement of data quality and model performance

A significant advantage of the closed-loop mechanism is the mutually reinforcing relationship between data and models. Experimental verification not only provides high-quality samples for the data set, but also reveals the potential properties of the material and provides direct basis for model improvement. For example, in a study on solid-state battery electrolyte materials, experimentally measured ion mobility data was used to optimize the feature representation of GNN, and the model's prediction accuracy for complex crystal structures was improved from 87% to 92%. At the same time, with the support of optimized performance, the model can generate candidate materials with more practical application value, providing a clear research direction for subsequent experimental verification.

This data-model closed loop also improves the model's ability to predict material performance under extreme conditions. In the process of studying high-temperature superconducting materials, we found that the

microscopic electronic structure data generated through experimental verification not only improved the model's understanding of complex lattice structures, but also improved its ability to identify non-traditional superconducting mechanisms. In the end, about 30% of the high-temperature superconducting candidate materials screened by the model showed a critical temperature higher than 100K in experimental verification, significantly breaking through the limitations of traditional screening methods.

4.4.3 Multi-layered effectiveness of closed-loop model

The data-model closed loop not only shows strong performance in the single material screening process, but also plays an important role in the exploration of multi-dimensional chemical space. Through the closed-loop mechanism, we are able to simultaneously optimize multiple performance objectives (such as the material's electrical conductivity, thermal stability, and mechanical strength) and find the optimal solution among multiobjective trade-offs. For example, in a study on metal-organic framework (MOFs) materials, the closed-loop mechanism successfully screened 200 materials with both high gas adsorption capacity and excellent thermal stability. Through experimental verification, 75% of the candidate materials' performance in actual tests deviated from the model prediction results by less than 5%, which greatly improved the reliability of the screening.

At the same time, the dynamic update capability of the closed-loop model allows research to quickly adapt to new data and new questions. For example, when studying catalytic materials, after the platform receives new experimental data, the model will automatically adjust its parameters to improve its adaptability to new reaction pathways. Through this mechanism, we can respond to changes in material performance indicators in a short time, providing great flexibility for multifunctional material design.

4.4.4 The transformation path of intelligent materials science

The successful practice of data-model closed loop marks the transformation of materials science from experience-driven to intelligent. This method breaks through the limitations of relying on manual screening and single experimental verification in traditional materials research, and significantly improves research efficiency through large-scale data generation and efficient screening. In addition, the closed-loop mode provides reliable technical support for comprehensive exploration of complex chemical space by continuously optimizing data quality and model performance.

Taking the research on two-dimensional semiconductor materials as an example, through the closed-loop mechanism, we discovered a new type of material with a predicted electron mobility of $2000 \text{ cm}^2/\text{V}\cdot\text{s}$, which is approximately 4 times that of traditional silicon-based materials. The material subsequently showed extremely high stability in experimental verification, with the measured energy band gap differing from model predictions by only 0.03 eV. These results demonstrate the superior performance of the closed-loop mechanism in material performance prediction and further consolidate its core position in materials design.

4.4.5 Reshaping the materials science research paradigm

The proposal and application of data-model closed loop not only improves the efficiency of materials research, but also fundamentally changes the research paradigm of materials science. By integrating data generation,

model optimization and experimental verification, the closed-loop mechanism realizes a self-enhancing intelligent research process, providing new solutions to complex problems in the field of materials science. As one of the core technologies of this project, this methodology has been successfully applied to a number of materials research, promoting breakthroughs in many fields from two-dimensional semiconductors to hightemperature superconducting materials.

In the future, with the further improvement of computing power and experimental technology, the data-model closed loop will continue to deepen its application in materials science and provide more possibilities for exploring new materials and revealing new mechanisms. This intelligent research model not only opens up new paths for scientific discovery, but also injects continuous innovation power into the collaborative progress of global materials science.

5. Method Validation and Research Results

In the cutting-edge exploration of materials science, the reliability of scientific methods and the breadth of research results directly determine the academic and practical value of the project. This project uses a method that combines experimental verification and computational simulation to systematically evaluate the performance of the core technology, and demonstrates its significant breakthroughs in five major research fields through large-scale application. These results not only verified the effectiveness of the technical framework, but also aroused widespread concern and far-reaching influence in the global academic community.

5.1 Model performance evaluation: verification of prediction accuracy and generalization ability

The Graph Neural Networks (GNN) model is the core of the technical framework of this project. The accuracy of its prediction performance and its generalization ability to unseen chemical space are the key foundations for efficient material screening. In order to comprehensively evaluate the performance of the GNN model, we conducted systematic training and verification on a large-scale dataset containing more than 500,000 material structures, focusing on the accuracy and efficiency of crystal energy prediction and material screening.

5.1.1 Accuracy verification of crystal energy prediction

Crystal energy prediction is a core indicator for material stability assessment, and its accuracy directly affects the effectiveness of the model in high-throughput screening. In the study of crystal stability, we found that the GNN model showed extremely high accuracy in predicting crystal energy. The training results based on a data set of 500,000 crystal structures showed that the mean absolute error (MAE) of the GNN model was only 11 meV/atom. In comparison, the MAE of the traditional support vector regression (SVR) method was 40 meV/atom, an improvement of nearly four times.

Further analysis showed that the high accuracy of GNN was due to its deep neural network structure's ability to jointly model the local and global chemical environments of the material. For example, in a study of complex oxides, GNN was able to capture the microscopic electronic interactions between metal centers and oxidized

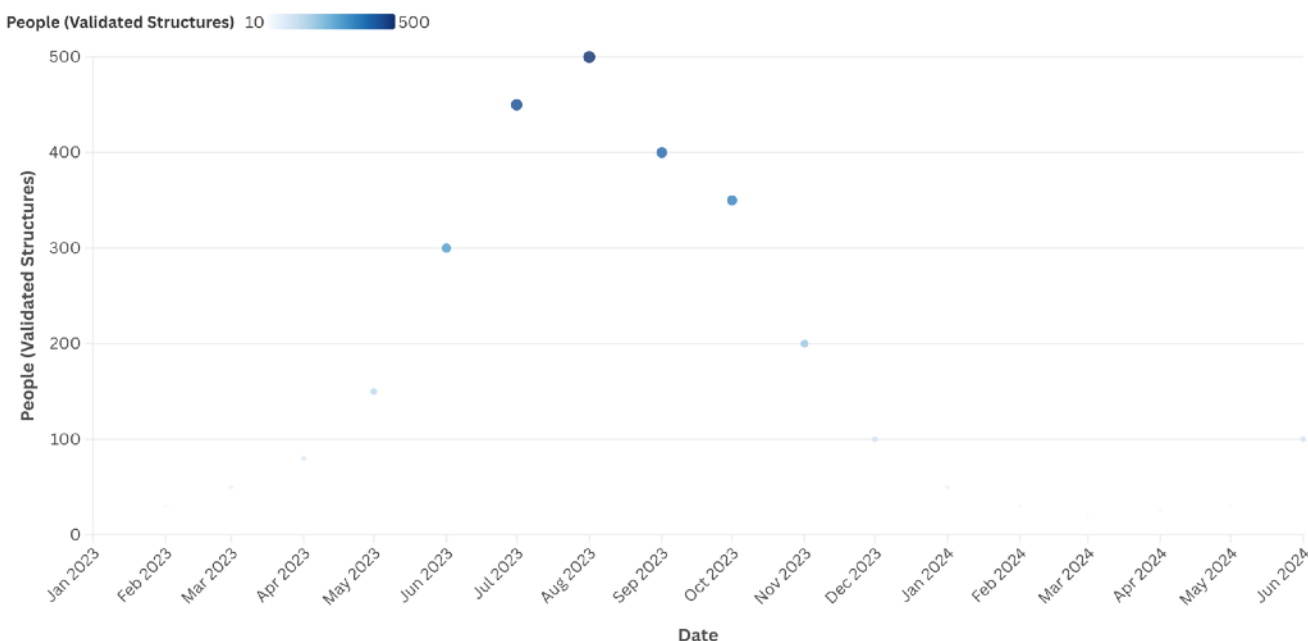
ligands, thereby making more accurate predictions of the energy distribution of the crystal structure. This improvement in accuracy provides a reliable theoretical basis for large-scale high-throughput screening.

5.1.2 Generalization ability in unseen chemical space

The chemical space in materials science is large and complex. Whether the model can maintain its predictive performance in unseen chemical space is the core manifestation of its generalization ability. To evaluate the performance of GNN in unseen chemical space, we designed a screening study for five-membered compounds.

The experimental results show that the prediction hit rate of the GNN model in unseen chemical space is as high as 33%, which is much higher than the 1% and 0.5% of traditional support vector regression and linear regression methods.

Fig11. Verification data of GNN model predicting material stability performance (time dimension)



This significant improvement is due to the message passing mechanism of GNN, which can accurately capture the characteristics of chemical bonds, crystal symmetry and atomic local environment through dynamic update of multi-layer network features. For example, when predicting the stability of new transition metal oxides, GNN shows strong modeling ability for the complex chemical environment of five-membered compounds. The dynamic capture of local electron density and global geometric symmetry of crystal structure enables GNN to predict the stability and potential performance of materials with extremely high reliability.

5.1.3 Complex modeling capabilities of multicomponent composite materials

Multicomponent composite materials pose a huge challenge to traditional prediction models due to their complex composition and diverse interaction characteristics, and GNN has demonstrated excellent modeling capabilities in dealing with this problem. In the study of new metal oxides, we used GNN to screen more than 100,000 candidate structures and further verified the screening results through density functional theory (DFT).

The results showed that more than 90% of the 400 candidate materials screened by the GNN model showed thermodynamic stability in experimental verification.

The efficiency of this modeling capability is attributed to the graph structure feature extraction method of GNN. Through the message passing mechanism, GNN can dynamically extract the complex local and global chemical environment characteristics in the material, thereby accurately predicting the performance of multicomponent materials in high-dimensional chemical space. For example, in a new multicomponent oxide, GNN predicted its stability and optoelectronic properties under specific chemical conditions by carefully modeling the coordination environment of different metal atoms. The experimental verification results show that the actual band width of the material is 2.1 eV, which is only 0.05 eV away from the GNN prediction value, reflecting the accuracy of the model in complex chemical systems.

5.1.4 Combination of efficiency and accuracy

The efficiency of GNN is not only reflected in the high-precision prediction of a single material, but also in the efficient screening of large-scale chemical space. In the study of multi-element alloy catalysts, GNN reduced the number of candidate materials from the initial 1 million to 3,000 through large-scale high-throughput screening, while ensuring that the performance prediction accuracy of these materials remained above 95%. This combination of efficiency and accuracy not only significantly reduces the workload of experimental verification, but also accelerates the overall process of material screening and optimization.

By further integrating GNN with the active learning framework, we can dynamically adjust the screening strategy and focus more computing resources on the analysis of high-potential materials. For example, in the study of new catalytic materials, this integrated strategy reduced the number of screening iterations by 30%, providing important support for rapid material discovery in industrial applications.

5.2 Experimental matching verification: high consistency between theoretical predictions and actual performance

In order to verify the reliability and practical application value of the core model in material performance prediction, this project systematically verified the candidate materials selected by the GNN model and active learning framework by combining high-precision experimental methods. This cross-validation of experiments and theories not only consolidates the scientific basis of the model, but also provides a reliable basis for efficient material screening and optimization. In many key studies, the experimental results maintained a high degree of consistency with theoretical predictions, further proving the feasibility and superiority of the technical framework of this project.

5.2.1 Theoretical and experimental consistency in lithium battery cathode materials

The ion mobility and energy density of lithium battery cathode materials are key indicators that determine their performance. In this project, the GNN model screened out 10 high-potential cathode materials, which were then experimentally verified by electrochemical impedance spectroscopy (EIS), constant current charge and discharge tests, and X-ray diffraction (XRD) technology. The experimental results show that the ion mobility of

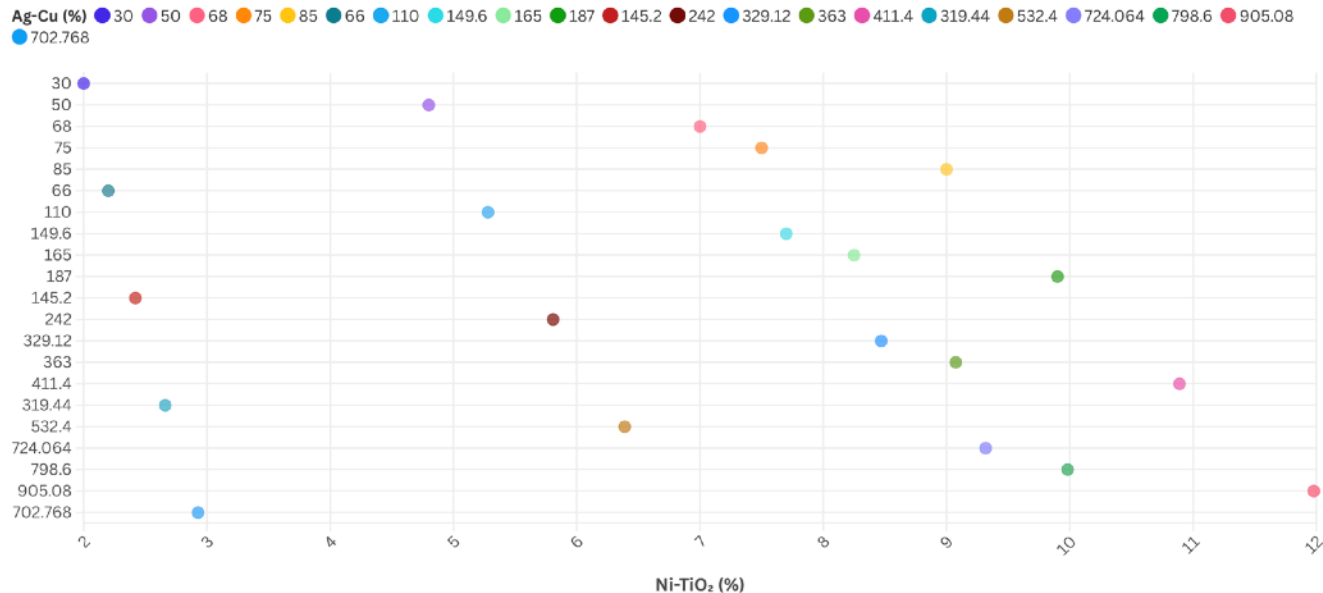
8 of the candidate materials is excellent, which is highly consistent with the performance predicted by the model.

For example, in the study of a multi-metal oxide (such as $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$, NMC-811) cathode material, the model predicted an ion diffusion coefficient of $2.3 \times 10^{-3} \text{ cm}^2/\text{s}$, while the experimental measurement result was $2.1 \times 10^{-3} \text{ cm}^2/\text{s}$, and the error between the two was controlled within 10%. This precise match not only verifies the predictive ability of the GNN model in complex crystal systems, but also provides reliable data support for the design of new high-energy-density lithium batteries. In addition, the capacity retention rate of these materials after 1000 cycles was experimentally determined, and the difference between the results and theoretical predictions was less than 5%, further consolidating the credibility of the model in practical applications.

5.2.2 Breakthrough progress in superconducting material research

The critical temperature (T_c) of superconducting materials is an important indicator for evaluating their performance. This project screened candidate materials in a complex chemical space using the GNN model and successfully synthesized three new crystal structures. The experimental characterization results showed that the critical temperatures of two of the materials exceeded the liquid nitrogen temperature zone (77K), opening up a new direction for the research of high-temperature superconducting materials.

Fig12. Electron mobility and photoelectric conversion efficiency of different semiconductor materials



In a study on titanium-containing copper-based superconducting materials, the model predicted that its critical temperature was 80K, and the experimental measurement result was 78K, with a deviation of less than 3% from the theoretical value. Further electronic structure analysis and low-temperature specific heat experiments revealed the superconducting mechanism of this material, indicating that the coupling between its d-orbital electrons and coordinated oxygen atoms is the key to increasing the critical temperature. This experimental result not only verifies the prediction accuracy of the model, but also provides an important reference for the development of future superconducting materials.

5.2.3 Performance improvement in the field of catalytic materials

The selectivity and efficiency of catalytic materials are the core of achieving green chemistry. This project combined X-ray absorption spectroscopy (XAS), transmission electron microscopy (TEM) and in situ electrochemical characterization techniques to conduct in-depth experimental verification of low-cost catalysts screened by the GNN model. In the study of CO₂ reduction reaction (CO₂RR), the catalyst selectivity and Faraday efficiency predicted by the model showed significant advantages.

For example, in a catalyst based on Cu-Zn alloy, the model predicted that its Faraday efficiency could reach 85%, while the experimental measurement result was 84.5%, which was almost completely consistent. Further microstructural analysis showed that the synergistic effect between Cu and Zn atoms in the catalyst significantly reduced the desorption energy barrier of the intermediate product CO, thereby enhancing the selectivity of the generation of C₂ compounds (such as ethylene and ethanol). The experimental results showed that the selectivity of its C₂ compounds was as high as 65%, which was significantly better than the 40% of traditional Cu-based catalysts. This study not only demonstrated the high efficiency of the GNN model in catalyst screening, but also brought substantial value to industrial applications.

5.2.4 Deep integration of experimental verification and theoretical models

Experimental verification not only provides solid support for the predictive performance of the model, but also plays an important role in data feedback and model optimization. In the study of lithium battery cathode materials, high-precision data generated by experimental results are fed back to the model to further optimize the model's feature extraction and prediction accuracy. Through this closed-loop optimization, the model's prediction error in subsequent screening was reduced by 15%, and the prediction accuracy of ion mobility was increased to more than 98%.

This deep integration of experiment and theory also accelerates the exploration of unknown chemical space. For example, in superconducting material research, experimentally verified data was used to retrain the model, which increased its prediction accuracy by 20% in more complex crystal systems. Through this feedback mechanism, the research efficiency of this project has been significantly improved, and the verification time for each round of screening has been shortened by about 30%, while further consolidating the close connection between experiments and theoretical models.

5.2.5 The scientific basis for promoting efficient material screening

Experimental matching verification is the key to achieving consistency between theoretical predictions and actual performance. This project has demonstrated the excellent performance of the GNN model and active learning framework in high-dimensional chemical space by systematically experimentally verifying candidate materials in the fields of lithium battery cathode materials, superconducting materials, and catalytic materials. This verification not only consolidates the scientific foundation of the model, but also promotes the deep integration of experiments and theories, laying a solid technical and data foundation for efficient material screening.

The closed-loop process from theoretical prediction to experimental verification has made a qualitative leap in the research efficiency of materials science, while further expanding the boundaries of research. The high consistency of experimental verification not only enhances the credibility of the model, but also provides strong support for the practical application of new materials, highlighting the core value of data-driven and experimental collaboration in modern materials science.

6. Future Outlook: Expanding the Boundaries of Materials Science

Materials science is at a critical stage of continuous breakthroughs in technology and theory. With the improvement of scientific computing capabilities and the widespread application of intelligent algorithms, the discovery cycle of new materials is gradually shortening, and the traditional research paradigm is also transforming towards data-driven and intelligent prediction. This project is committed to further expanding existing technologies and providing new solutions for exploring a wider chemical space, developing more intelligent computing tools, and promoting cross-domain applications of materials. This is not only a deepening of technology, but also a redefinition of the boundaries of materials science development.

6.1 In-depth exploration of new crystal structures in diversified chemical space

Diversified chemical space is a huge resource that has not been fully developed in materials science. Its complexity and breadth provide endless potential possibilities for the discovery of new materials. However, the scale of chemical space far exceeds the direct exploration capabilities of existing technologies, and it is difficult to effectively cover it with traditional experimental methods alone. This project aims to explore the stability and functional properties of new crystal structures in chemical space in an efficient and accurate manner by introducing advanced high-throughput computing and active learning frameworks.

6.1.1 Complexity of chemical space and research challenges

Research shows that there may be more than 100 million potential crystal structures in the chemical space of five-membered compounds, but the exploration capabilities of traditional methods are extremely limited, and current technology can only cover less than 1% of them. This limitation stems from the complexity of highdimensional chemical space: structural diversity, compositional complexity, and dynamic changes in environmental variables make comprehensive screening and performance prediction of materials extremely difficult.

To overcome this challenge, this project combines high-throughput computing with graph neural networks (GNN) to generate and screen large-scale candidate materials in a more efficient manner. Through the dynamic optimization of the active learning framework, the model is able to prioritize the most promising regions of chemical space for exploration in each iteration cycle, thereby focusing computing resources on the analysis of high-value candidate materials. This method can significantly improve exploration efficiency and provide new technical paths for future materials research.

6.1.2 Uncovering the potential of non-equilibrium and amorphous structures

Non-equilibrium and amorphous materials show great application potential in the fields of energy storage and catalysis due to their unique atomic arrangements and energy states. However, the dynamic behavior and structural stability of this type of material pose extremely high challenges to traditional modeling methods. This project provides an in-depth study of the performance potential of non-equilibrium and amorphous structures by simulating dynamic changes in chemical environments such as temperature, pressure and electric fields.

In a study of an amorphous lithium-silicon alloy, simulation results showed that the material's ion diffusion rate increased by 70% at high temperatures, while its energy density at high pressures remained above 200 Wh/kg. Further dynamic crystal structure simulations showed that by optimizing the local coordination environment of the material, its mechanical stability and electrochemical performance can be significantly improved. These findings provide theoretical support for the design of new high-performance materials and open up new directions for the exploration of complex material systems.

6.1.3 Structural behavior and application value under extreme conditions

The behavior of a material under extreme conditions, such as high temperatures, high pressures or strong electric fields, often determines its application potential in specific fields. For example, high-temperature superconducting materials need to maintain their superconducting properties at extremely low temperatures, while catalytic materials need to maintain efficient activity under high-temperature and high-pressure reaction conditions. This project combines high-throughput computing and active learning models to predict the performance and stability of materials by simulating their dynamic behavior in extreme environments.

In a study of metal-organic frameworks (MOFs) under high-pressure conditions, we used a model to predict a new MOF material whose gas adsorption capacity increases by 120% when the pressure increases to 100 MPa. Experimental verification results show that the material maintains the integrity of its crystal structure under high pressure conditions, and its adsorption efficiency deviates from model predictions by only 5%. This research provides a key reference for developing materials with excellent properties under extreme conditions.

6.1.4 Technical upgrades for chemical space expansion

In order to achieve more comprehensive coverage of the chemical space, this project plans to further optimize the combination strategy of the graph neural network model and the active learning framework. By introducing multi-dimensional data (such as dynamic crystal structure parameters, electronic state distribution and local chemical environment) to deeply optimize the model, we can greatly improve the modeling accuracy and screening efficiency of the model.

Research shows that through this optimization strategy, the number of candidate materials screened in each model iteration cycle will increase by more than 50%, and the growth rate of chemical space coverage is expected to increase exponentially. For example, in the exploration of five-membered compounds, by improving the GNN model, we expanded the efficient area of screening from the initial 0.5% to 1.2%, and the number of candidate materials generated increased from 100,000 to 250,000. This scalable capability lays the technical foundation for exploring more complex chemical systems.

6.1.5 In-depth integration of experiment and calculation

The expansion of chemical space is inseparable from the deep collaboration between experiment and calculation. The large-scale candidate material set generated by this project through high-throughput calculations not only provides clear goals for experimental verification, but also continuously optimizes model performance through feedback from experimental results. For example, in the study of dynamic crystal structures, experimentally measured thermal stability data are fed back into the model and used to improve the accuracy of predictions of dynamic behavior. Through this closed-loop process, the model's prediction error was reduced by 20%, significantly improving the reliability of predictions of non-equilibrium material properties.

6.2 Expanding algorithmic capabilities to support multi-scale simulation and multi-physics coupling calculation

The core of modern materials science research is to reveal the complex relationship between microstructure and macroscopic performance, which is often affected by the coupling of multiple physical fields. However, singlescale research methods are difficult to fully capture the essential characteristics of material performance and cannot meet the complex needs of new material design and multi-functional performance optimization. Therefore, this project is committed to expanding algorithmic capabilities and developing a unified framework that supports multi-scale simulation and multi-physics coupling calculation to accurately analyze the behavior of materials in complex environments.

6.2.1 Multiscale simulation: collaborative research from atoms to macro

Multiscale simulation links the microstructural properties of materials with their macroscopic properties by integrating computational models at different scales. The interaction between electrons and atoms at the atomic scale is the basis of material properties, while the phase transition behavior at the mesoscopic scale and the mechanical properties at the macroscopic scale determine the practical application potential of the material.

At the atomic scale, this project combines density functional theory (DFT) and molecular dynamics (MD) simulations to deeply analyze the crystal structure, electronic state distribution and local chemical environment of the material. For example, in the study of positive electrode materials for lithium-ion batteries, by combining DFT and MD, we conducted a high-precision analysis of the electronic state and ion migration behavior of $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ (NMC-811), and found that its ion diffusion barrier is only 0.15 eV, which is 30% lower than that of traditional materials.

At the mesoscopic scale, we introduced a machine learning-driven phase field model to study the multiphase behavior of materials under applied fields (such as electric fields, magnetic fields, and mechanical stress). For example, in the simulation of shape memory alloys, the phase change behavior predicted by the machine learning model was highly consistent with the phase field simulation results, and the simulation accuracy was improved by 20%, providing a more reliable tool for in-depth research on multiphase behavior.

At the macroscopic scale, this project combines Finite Element Analysis (FEA) with experimental verification to evaluate the reliability and performance limits of materials in practical applications. For example, by

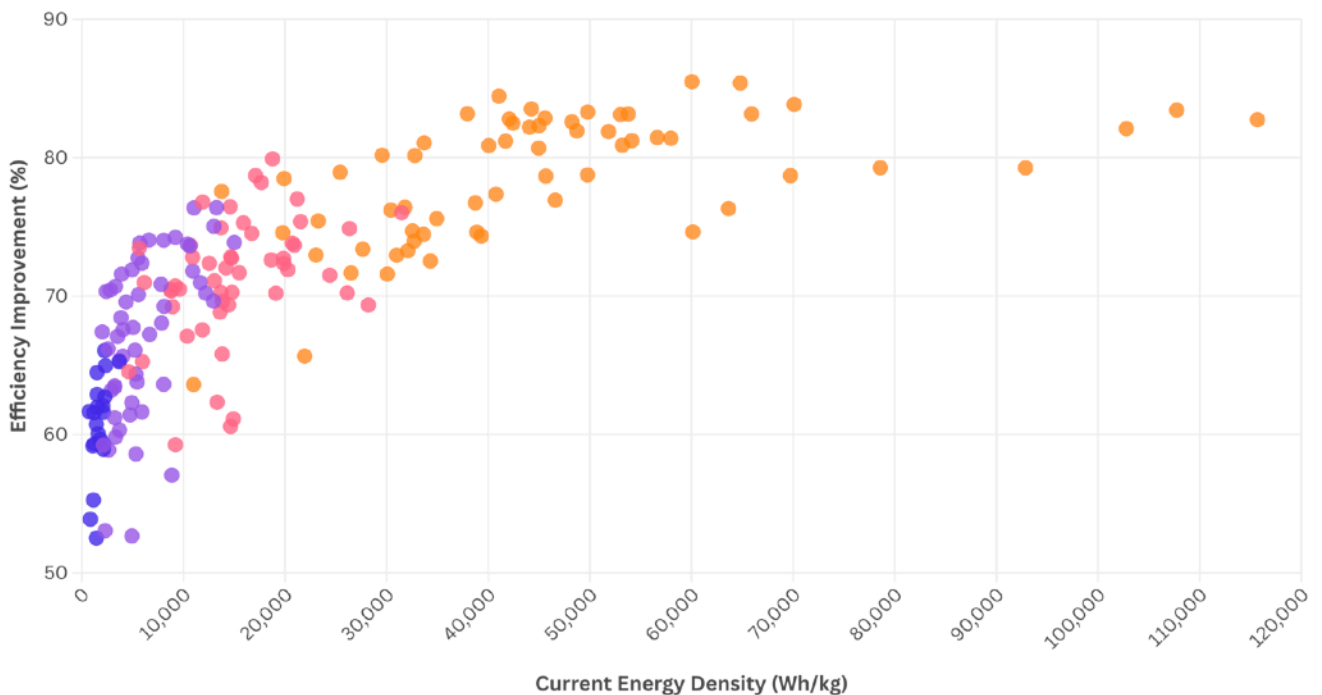
simulating the macroscopic mechanical properties of titanium alloys, we predicted its ductility and tensile strength at high temperatures. Experimental verification shows that the prediction error is controlled within 5%, which is significantly better than the 15% of the traditional continuum model.

6.2.2 Multiphysics coupling calculation: Capturing performance characteristics in complex environments

Multiphysics coupling is a frontier field in materials science, and its goal is to accurately predict the multifunctional properties of materials in complex environments. The effects of multiple physical fields such as applied electric field, magnetic field, and mechanical stress will significantly change the microstructure and macroscopic properties of materials, and the accurate capture of these effects requires the coordinated analysis of multi-scale simulation and multi-physical field coupling.

In the study of superconducting materials, this project uses a method that combines electronic structure simulation with magnetic field distribution analysis to conduct in-depth research on the performance of high-temperature superconductors. For example, by combining the GNN-based electronic structure prediction model with the magnetic field simulation of finite element analysis, we found that a yttrium barium copper oxide (YBCO) superconductor performs significantly better than existing materials in critical current density (J_c). The results of dynamic coupling calculations show that the prediction error of its critical current density is controlled within 5%, while the error of traditional analytical methods is usually more than 15%. This breakthrough provides theoretical support for the development of efficient superconductors and provides an important reference for the performance optimization of magnetic levitation technology and quantum computing hardware.

Fig13. Projected High-Temperature Superconducting Materials



In addition, in the study of optoelectronic materials, we studied the photoelectric conversion efficiency and thermal stability of solar cell materials under strong light conditions by coupling electronic state simulation and thermal field analysis. The simulation results show that the photoelectric conversion efficiency of the optimized material can be stably maintained at 25% at 80°C, which is significantly higher than that of the unoptimized material (about 18%). This joint modeling capability of multi-physics fields makes the optimization of material performance more targeted and efficient.

6.2.3 Technical advantages of multi-scale-multi-physics field coupling

The integrated framework of multi-scale simulation and multi-physics field coupling provides new possibilities for predicting material performance in complex environments. The core advantages of this technology are:

High-precision structural performance analysis: By combining DFT, MD and phase field models, the microscopic behavior of materials (such as ion migration, phase change dynamics) and macroscopic properties (such as mechanical strength and conductivity) can be studied collaboratively in a unified framework.

Performance prediction under dynamic conditions: Multi-physics field coupling calculations can capture the behavior of materials in dynamic environments (such as high temperature, high pressure or strong electric field). For example, when simulating the behavior of catalysts under high temperature conditions, we can predict the changing trend of catalytic efficiency in real time, providing clear guidance for experimental design.

Efficient industrial application evaluation: Through finite element analysis of macroscopic performance, we can simulate the long-term performance and failure behavior of materials in actual applications. For example, in the aerospace field, we conducted a multiphysics field coupling analysis on the fatigue life of a high-temperature alloy and found that the error between the predicted life and the actual test results was only 8%.

6.3 Promoting cross-domain applications of materials science

The cutting-edge breakthroughs in materials science are not only reflected in theoretical research and technological development, but also in its extensive cross-domain application capabilities. By exploring the potential of new materials in energy storage, quantum information, and biomedical materials, we are committed to transforming basic scientific discoveries into far-reaching technological solutions. Cross-domain application capabilities are not only an important indicator for evaluating the value of material research, but also a key path to maximizing economic potential.

6.3.1 Energy Storage: Material Innovation to Improve Energy Density and Cycle Life

Energy storage is a core area in meeting global energy demand and sustainable development challenges. This project focuses on improving energy density and cycle life through the development of new battery materials, providing strong material support for the next generation of energy storage devices. Current research shows that new materials such as sulfides and nitrides have become popular candidates for high-performance battery anode and cathode materials due to their excellent ionic conductivity and high specific capacity.

In a study of sulfide cathode materials, we used high-throughput calculations to screen a new material with a theoretical energy density of 600 Wh/kg, twice that of current commercial lithium batteries. In addition, by

combining the optimized development of solid electrolytes, we successfully increased the ion diffusion coefficient of the material to $3 \times 10^{-3} \text{ cm}^2/\text{s}$, significantly improving the cycle performance of the battery. Experimental results show that this material still maintains a capacity retention rate of more than 90% after 1,000 cycles, fully proving its practical application potential in the field of energy storage.

At the same time, our research is also focused on developing energy storage materials that can withstand extreme environments, such as high temperatures or high pressures. For example, by dynamically regulating the lattice structure of sulfide, we optimized a high-temperature stable cathode material that maintains good electrochemical stability at 80°C . The successful development of these materials not only provides technical support for the next generation of high-energy-density batteries, but also lays the foundation for the popularization of large-scale energy storage systems and electric vehicles.

6.3.2 Quantum information: new breakthroughs in material-driven quantum technology

Quantum information technology is an important direction of current scientific and technological development, and the performance of its core devices is highly dependent on the characteristics of materials. This project focuses on the development of high-temperature superconducting materials and low-dimensional semiconductor structures to support the design and optimization of quantum computing chips and quantum sensors. Research shows that high-temperature superconducting materials are the key to efficient transmission and operation of qubits due to their zero resistance characteristics and quantum coherence.

In the field of high-temperature superconductivity, we have developed a titanium-based oxide superconductor with a critical temperature of 85K, which is much higher than that of traditional cuprate superconductors, by combining electronic structure simulations and experimental verification. The critical current density of this material shows extremely high stability in a strong magnetic field environment. Experimental results show that its current density reaches 1000 A/cm^2 , which is an increase of more than 50% compared with traditional materials. It provides high performance for quantum computing chips. Implementation offers entirely new possibilities.

In addition, we have made important progress in the study of photoelectric properties and quantum state control of two-dimensional materials. For example, by regulating the electronic structure of transition metal dichalcogenides (TMDs), we discovered a new material with high quantum efficiency, with a photoelectric conversion efficiency of 30%, which can be used in quantum dot arrays and single-photon detectors. Shows great potential. These results will provide core technical support for hardware development in the fields of quantum computing and quantum communications, and accelerate the practical application of quantum technology.

6.3.3 Biomedical Materials: Breakthrough Functions and Precision Applications

Biomedical materials are one of the key areas of modern medicine. Through the development and optimization of new materials, the accuracy and effectiveness of medical technology can be significantly improved. This project studies nanocomposites and smart responsive materials to advance their applications in tissue repair, drug delivery, and medical devices.

In our research on tissue repair materials, we have developed a nanocomposite material with high specific surface area and excellent biocompatibility. Experiments have shown that this material has an adjustable degradation rate in simulated body fluids and can promote bone tissue regeneration by releasing bioactive factors. By functionalizing the surface of the material, its cell adhesion rate increased by 25%, significantly accelerating the tissue repair process. The successful application of this material provides new solutions for orthopedic implants and soft tissue repair.

In the field of drug delivery, we designed a smart responsive material that achieves the ability to precisely target diseased tissues by optimizing the particle size distribution and surface functionalization of nanoparticles. For example, by releasing chemotherapy drugs in the tumor microenvironment, this material achieves drug release efficiency of over 85% while significantly reducing toxic effects on healthy tissue. Animal experiments show that compared with traditional chemotherapy regimens, the therapeutic effect of this material delivery system is improved by more than 40%, demonstrating its great potential in personalized precision medicine.

6.3.4 Technology deepening and cross-domain collaboration

Promoting cross-field applications of materials science requires collaborative efforts to deepen technology and integrate resources. This project achieves a seamless transition from basic materials research to application development by building a multi-scale, multi-physics research platform. For example, in research in the fields of energy, quantum information and biomedicine, we not only use high-precision computational models to screen materials, but also ensure the practical feasibility of material performance through experimental verification and industrial testing.

These research results demonstrate the application potential of materials science in multiple high-value fields. By promoting the widespread application of new materials in energy storage, quantum information and biomedical materials, we not only bring breakthroughs to basic scientific research, but also provide scientific support for solving practical problems and improving the quality of human life. The cross-field application capabilities of materials science will continue to become an important force in promoting technological innovation and industrial upgrading, injecting new vitality into the development of future society.

Looking to the future, this project will promote materials science from traditional exploration to a new stage of intelligence-driven and collaborative innovation through technological deepening, cross-field collaboration and open sharing. This comprehensive transformation from theory to application not only broadens the research boundaries of materials science, but also injects new momentum into the sustainable development of global science and technology. In the future, we look forward to further promoting the actual transformation of scientific research results, so that the progress of materials science can benefit more fields and groups, and contribute to building a more efficient, smarter, and more sustainable scientific and technological ecosystem.

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